وزارة الصحة العراقية
Iraqi Ministry of Health
Founded 1920 تأسست

دائرة الصحة العامة
Public Health Directorate

2025

VOLUME 1

# NATIONAL GUIDELINES FOR PRIMARY HEALTH CARE PHYSICIANS

## Chronic Respiratory Diseases.(CRDs)

# Management of Asthma



Directorate of Public Health
Non-Communicable Disease
Department

IRAQ NCD

وزارة الصحة العراقية
Iraqi Ministry of Health
Founded 1920 تأسست

دائرة الصحة العامة
Public Health Directorate

# National Guidelines for Primary Health Care Physicians

# Guidelines for the Management of Asthma for Primary Health Care


Directorate of Public Health

Non-Communicable Disease Department



2025

## Supervision and revision by:

- Dr. Riyadh Abdul Ameer AL-Hilfy, DG of the Directorate of Public Health, MOH.
- Dr. Muna Atallah Khaleefa Ali, Consultant Community Medicine/ Director of the Noncommunicable prevention and control Department, Directorate of public health, MOH.

## Contributors to the National Guideline for Asthma Management version 3

- Dr. Basil Fawzi Jameel FIBMS /internal medicine / FIBMS /respiratory medicine/ consultant of respiratory medicine Baghdad teaching hospital / medical city.
- Dr. Ola Shakir Fadhil / FICMS / Family Medicine / Directorate of Public Health / NCD department.
- Dr. Nada Abdul Wahhab Mousa / Consultant Community Medicine / Manger of Compressive management of NCDs section / NCD department / Directorate of Public Health, MOH
- Dr.Hind Mhamood kamal ( M.D.F.I.C.M.S) / Community Medicine / Directorate of Public Health / NCD department.
- Abdulrahman I. Hussein / Optometrist / Eye Care Unite, Non-Communicable Prevention & Control Department, Directorate of Public Health, MOH.
- Marwan Faisal Mohsin /statistician / CRDs unit / Directorate of Public Health / NCD department / MOH .

## Contributors to the National Guideline for Asthma Management: version 2

- Dr. Basil Fawzi Jameel FIBMS (internal medicine) FIBMS (respiratory medicine) consultant of respiratory medicine Baghdad teaching hospital medical city.
- Dr. Muna Attallah Khaleefah Ali, FICMS/ Community Medicine, Director of Non Communicable Department / MOH.
- Dr. Ola Shakir Fadhil / FICMS / Family Medicine / Directorate of Public Health / NCD department.
- Dr. Nada Abdul Wahhab Mousa / PhD in Community Medicine Directorate of Public Health, NCD department.

## Contributors to the National Guideline for Asthma Management: version 1

- Dr.Abdul Hameed Abdul Majeed , Specialist Doctor.
- Dr.Mustafa Abdul Fatah ,Specialist Doctor
- Dr. Faris Abdul Kareem Khazzal, Assistant Professor, Al Kindi Medical College.

- Dr. Muna Attallah Khaleefah Ali, FICMS/ Community Medicine, Director of Non Communicable Section / MOH.
- Dr. Nada Abdul Wahhab Mousa / PhD in Community Medicine, Primary Health Care Department, Directorate of Public Health , NCD Section.
- Dr. Bushra Ibrahim Abd Al Lattif / MSc in Community Medicine, Primary Health Care Department, Directorate of Public Health , NCD Section.
- Dr. Nada Abdul Wahhab Mousa / Consultant Community Medicine / Manger of Compressive management of NCDs section / NCD department / Directorate of Public Health, MOH.

| Abbreviation | Terminology |
|---|---|
| **AAP** | Asthma Action Plan |
| **ABPA** | Allergic Bronchopulmonary Aspergillosis |
| **ACE** | Angiotensin Converting Enzyme |
| **ACT** | Asthma Control Test |
| **AERD** | Aspirin-Exacerbated Respiratory Disease |
| **AIA** | Aspirin-Induced Asthma |
| **AIR** | Anti- Inflammatory Reliever |
| **COPD** | Chronic Obstructive Pulmonary Disease |
| **COVID-19** | Coronavirus Disease |
| **CRDs** | Chronic Respiratory Diseases |
| **EIB** | Exercise-Induced Bronchoconstriction |
| **EMRO** | Regional Office for The Eastern Mediterranean |
| **FEV1** | Forced Expiratory Volume in One Second |
| **FVC** | Forced Vital Capacity |
| **GAN** | The Global Asthma Network |
| **GERD** | Gastro Esophageal Reflux Disease |
| **GERD** | Gastroesophageal Reflux Disease |
| **GINA** | Global Initiative for Asthma |
| **ICS** | Inhaled Corticosteroids |
| **IgE** | Immunoglobulin E |
| **ISAAC** | International Study Of Asthma And Allergies In Childhood |
| **LABA** | Long Acting B2 – Agonist |
| **LAMA** | Long-Acting Muscarinic Antagonist |
| **LTRA** | Leukotriene Receptor Antagonist |
| **MART** | Maintenance-And-Reliever Therapy |
| **OCS** | Oral Corticosteroids |
| **PEF** | Peak Expiratory Flow |
| **PEFM** | Peak Expiratory Flow Meter |
| **RSV** | Respiratory Syncytial Virus |
| **SABA** | Short Acting B2 – Agonist |
| **WHO** | World Health Organization |

# Table of contents


1 Chronic respiratory diseases

2 Asthma
introduction, Definition of Asthma, Pathophysiology, Making the initial diagnosis

4 History and family history, Physical examination, Investigations

9 How to make the diagnosis of asthma in other contexts.

12 Assessment of asthma in adults, adolescents and children 6–11 years

17 Principles of asthma management in adults, adolescents and children 6–11 years

21 Non-pharmacological strategies

24 Medications and strategies for adults, adolescents, and children 6–11 years

26 Adults and adolescents: Asthma treatment tracks

29 About Asthma treatment for children 6-11 years

30 Guided asthma self-management education and skills training.

33 Managing asthma with multimorbidity and in specific populations

42 Management of worsening asthma and exacerbations in adults adolescents and children 6–11 years

45 References

46 Annexes

# Chronic Respiratory Diseases

Chronic respiratory diseases (CRDs) affect the airways and other structures of the lungs. Some of the most common are chronic obstructive pulmonary disease (COPD), asthma, occupational lung diseases and pulmonary hypertension. Over 3 million deaths every year are due to chronic respiratory diseases.

The two most common chronic respiratory diseases are asthma and chronic obstructive pulmonary disease (COPD). These both affect the airways in the lungs.

In asthma, the airway obstruction is reversible with inhaled medicines, but in COPD it is mostly fixed. COPD only affects adults and usually becomes worse with time.

Risk factors for chronic respiratory diseases include tobacco smoking (including second-hand smoke), air pollution, allergens and occupational risks. Outdoor air pollution and indoor air pollution (often caused by cooking with solid fuels) are also common causes. Asthma and COPD may be prevented by reducing or avoiding exposure to these risk factors.

Neither asthma nor chronic obstructive pulmonary disease (COPD) can be cured, however, various forms of treatment that help open the air passages and improve shortness of breath can help control symptoms and improve daily life for people living with these conditions. Treatment can reduce symptoms, prevent deterioration and improve daily life.

Appropriate management of asthma with inhaled corticosteroid medicine can control the progression of the disease and reduce deaths. Long-term treatment is required for people with persistent symptoms and exposures which trigger symptoms (e.g. smoke, fumes, dust, grass and tree pollen, animal fur and feathers) should be avoided.

Chronic obstructive pulmonary disease is confirmed by a spirometry test, which measures how much and how quickly a person can blow air out of their lungs. Symptoms can be treated through medical and physical treatments; however, with COPD in people who smoke tobacco, the most effective treatment available is to stop smoking. This can slow down the progression of the disease and decrease COPD-related deaths.

# Asthma

## Introduction

Asthma is a serious global health problem, affecting approximately 300 million people around the world, and causing around 1,000 deaths per day. It is often underdiagnosed and under-treated, particularly in low- and middle-income countries. People with under-treated asthma can suffer sleep disturbance, tiredness during the day, and poor concentration. Asthma sufferers and their families may miss school and work, with a financial impact on the family and wider community. If symptoms are severe, people with asthma may need to receive emergency health care and they may be admitted to a hospital for treatment and monitoring. Asthma continues to be a major source of global economic burden in terms of both direct and indirect costs.

Overall prevalence of current asthma symptoms in The Global Asthma Network (GAN) was found that asthma symptoms are present in 9.1% of children, 11.0% of adolescents, and 6.6% of adults. According to the World Health Organization an estimated adult asthma prevalence of 8% in the Eastern Mediterranean Region (EMRO). The prevalence of asthma appears to be higher in Saudi Arabia, with estimates reaching 14.2% in adults. In Syria, studies have shown a wheeze prevalence of 6.5% among adolescents (13-14 years old). Sudan has reported even higher figures, with studies suggesting that ever having asthma affects 15.5% and 18.2%.

Iraq's National Stepwise Non-Communicable Disease Risk Factor Survey conducted in 2015 found that 2.2% of the population reported ever having been diagnosed with asthma based on their medical history. This figure breaks down to 1.8% for men and 2.7% for women.

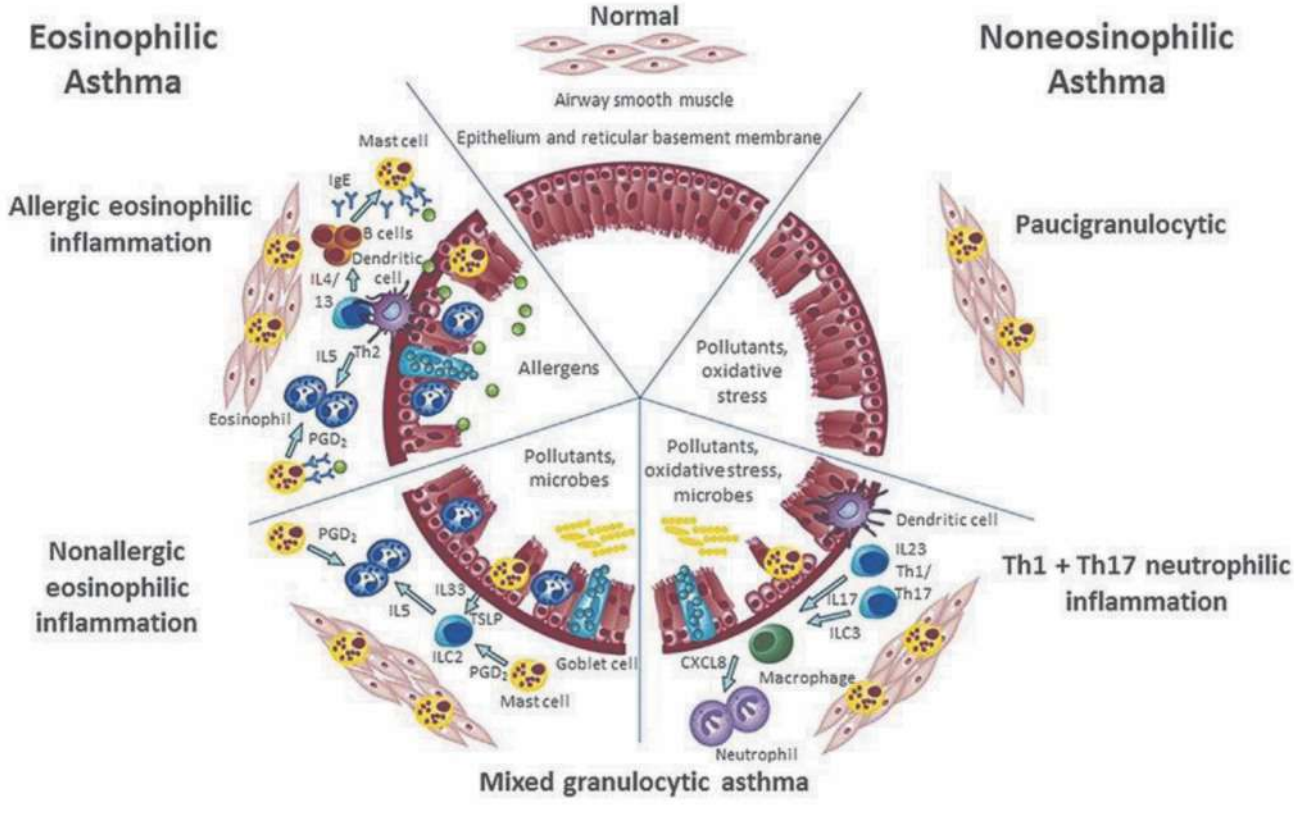
## Definition of Asthma

Asthma is a heterogeneous disease, usually characterized by chronic airway inflammation. It is defined by the history of respiratory symptoms, such as wheeze, shortness of breath, chest tightness and cough, that vary over time and in intensity, together with variable expiratory airflow limitation. One or more symptoms (e.g., cough) may predominate. Airflow limitation may later become persistent.

## Pathophysiology

Asthma is a chronic inflammatory airway disease that results in narrow airway lumen. The airway narrowing is caused by smooth muscle contraction, bronchospasm, and increased mucus secretion as well as bronchial wall thickening due to edema, smooth muscle hypertrophy and subepithelial fibrosis. The pathophysiological mechanisms that underlie these changes are diverse and heterogeneous. They are driven by variety of cell types including immune cells; mainly T-helper cells (Th2, Th17, Th1), B-cells, mast cells, eosinophils, dendritic cells, and neutrophils; as well as structural bronchial cells such as epithelial cells, myofibroblasts, and smooth muscle cells.

**Pathophysiology of Asthma**




## Making the initial diagnosis

Making the diagnosis of asthma before treatment is started is based on identifying both a characteristic pattern of respiratory symptoms such as wheezing, shortness of breath (dyspnea), chest tightness or cough, and variable expiratory airflow limitation. Globally, many health professionals lack access (or ready access) to spirometry, so advice has also been provided for using PEF in asthma diagnosis.

### Why is it important to confirm the diagnosis of asthma?

This is important to avoid unnecessary treatment or over-treatment, and to avoid missing other important diagnoses.

## History and family history

Commencement of respiratory symptoms in childhood, a history of allergic rhinitis or eczema, or a family history of asthma or allergy, increases the probability that the respiratory symptoms are due to asthma. However, these features are not specific for asthma and are not seen in all asthma phenotypes. Patients with allergic rhinitis or atopic dermatitis should be asked specifically about respiratory symptoms.

## Patterns of respiratory symptoms that are characteristic of asthma

The following features are typical of asthma and, if present, increase the probability that the patient has asthma. Respiratory symptoms of wheeze, shortness of breath, cough and/or chest tightness:

- Symptoms are often worse at night or in the early morning.
- Symptoms vary over time and in intensity.
- Symptoms are triggered by viral infections (colds), exercise, allergen exposure, changes in weather, laughter, or irritants such as car exhaust fumes, smoke, or strong smells.

The following features **decrease the probability** that respiratory symptoms are due to asthma:

- Chronic production of sputum
- Isolated cough with no other respiratory symptoms.
- Shortness of breath associated with dizziness, light-headedness, or peripheral tingling (paresthesia)
- Chest pain
- Exercise-induced dyspnea with noisy inspiration. (stridor)

## Physical examination

Physical examination in people with asthma is often normal. The most frequent abnormality is expiratory wheezing (rhonchi) on auscultation, but this may be absent or only heard on forced respiration. Wheezing may also be absent during severe asthma exacerbations, due to severely reduced airflow (so-called 'silent chest'), but at such times, other physical signs of respiratory failure are usually present. Wheezing may also be heard with inducible laryngeal obstruction, chronic obstructive pulmonary disease (COPD), respiratory infections, tracheomalacia, or inhaled foreign body. Crackles (crepitations) and inspiratory wheezing are not features of asthma. Examination of the nose may reveal signs of allergic rhinitis or nasal polyps.

## Investigations:

### Lung function testing to document variable expiratory airflow limitation

If possible, evidence of variable expiratory airflow limitation should be documented before treatment is started. This is because variability usually decreases with ICS treatment as lung function improves. In addition, any increase in lung function after initiating ICS-containing treatment can help to confirm the diagnosis of asthma. Bronchodilator responsiveness may not be present between symptoms, during viral infections or if the patient has used a beta2 agonist within the previous few hours; and in some patients with asthma, airflow limitation may become persistent or irreversible over time.

Peak expiratory flow (PEF) is less reliable than spirometry, but it is better than having no objective measurement of lung function. When measuring PEF, use the same meter each time as the value may vary by up to 20% between different meters, and use only the highest of three readings. divided by (mean of day's highest and lowest), averaged over two weeks. BD responsiveness may be lost temporarily during severe exacerbations or viral infections, and airflow limitation may become persistent over time.

If reversibility is not present at initial presentation, the next step depends on the availability of other tests and the urgency of the need for treatment. In a situation of clinical urgency, asthma treatment may be commenced and diagnostic testing arranged within the next few weeks, but other conditions that can mimic asthma should be considered, and the diagnosis confirmed as soon as possible.

**Box 1 Criteria for initial diagnosis of asthma in adults, adolescents, and children 6–11 years**

| **1. History of typical variable respiratory symptoms** | |
|---|---|
| ***Feature*** | ***Symptoms or features that support the diagnosis of asthma*** |
| **Wheeze, shortness of breath, chest tightnessand/or cough** (Descriptors may vary betweencultures and by age) | • Symptoms occur variably over time and vary in intensity<br>• Symptoms are often worse at night or on waking<br>• Symptoms are often triggered by exercise, laughter, allergens, cold air<br>• Symptoms often appear or worsen with viral infections |
| **2. Confirmed variable expiratory airflow limitation** | |
| **Feature** | ***Considerations, definitions, criteria*** |
| Excessive variability in expiratory lung function (one or more of the following): | The greater the variations, or the more occasions excess variation is seen, the more confident the diagnosis of asthma. If initially negative, tests can be repeated during symptoms or in the early morning. If spirometry is not possible, PEF* may be used, but it is less reliable. |
| Positive bronchodilator (BD) responsiveness (reversibility) test with spirometry (or PEF*) | Adults: increase from baseline in FEV1 or FVC of ≥12% and ≥200 mL, with greater confidence if the increase is ≥15% and ≥400 mL; or increase in PEF ≥20% if spirometry is not available.<br>Children: increase from baseline in FEV1 of ≥12% predicted (or in PEF of ≥15%).<br>Measure change 10–15 minutes after 200–400 mcg salbutamol or equivalent, compared with pre-BD readings. Positive test more likely if BD withheld before test: SABA ≥4 hours, long-acting bronchodilators 24–48 hours |
| Excessive variability in twice-daily PEF over 2 weeks | ***Adults***: average daily diurnal PEF variability >10%*<br>***Children***: average daily diurnal PEF variability >13%* |
| Increase in lung function after4 weeks of treatment | ***Adults***: increase from baseline in FEV1 by ≥12% and ≥200 mL (or PEF† by ≥20%) after 4 weeks of daily ICS-containing treatment<br>***Children***: increase from baseline in FEV1 of ≥12% predicted (or in PEF† of ≥15%). |
| Excessive variation in lung function between visits (good specificity but poor sensitivity) | ***Adults***: variation in FEV1 of ≥12% and ≥200 mL (or in PEF† of ≥20%) between visits.<br>***Children***: variation in FEV1 of ≥12% in FEV1 (or ≥15% in PEF†) between visits |

Withholding periods: Short-acting beta2 agonists: ≥4 hours; formoterol, salmeterol: 24 hours; indacaterol, vilanterol: 36 hours; tiotropium, umeclidinium, aclidinium, glycopyrronium: 36–48 hours.

## Confirming the diagnosis of asthma in patients already taking ICS-containing treatment

If the basis of a patient's diagnosis of asthma has not previously been documented, confirmation with objective testing should be sought. If the diagnosis of asthma cannot be confirmed, refer the patient for expert investigation and diagnosis.

**Box 2 Steps for confirming the diagnosis of asthma in a patient already taking ICS-containing**

| Current status | Steps to confirm the diagnosis of asthma |
|---|---|
| Variable respiratory symptoms and variable airflow limitation | **Diagnosis of asthma is confirmed.** Assess the level of asthma control and review ICS-containing treatment. |
| Variable respiratory symptoms but no variableairflow limitation | **Consider repeating spirometry** (or PEF*) after withholding bronchodilator (4 hrs for SABA, 24–48 hrs for long-acting bronchodilators or during symptoms. Checkbetween-visit variability of FEV1, and bronchodilator responsiveness. If still normal, consider other diagnoses.<br>*If FEV1 (or PEF*) is >70% predicted*: consider stepping down ICS-containing treatment and reassess in 2–4 weeks, then consider bronchial provocation testor repeating bronchodilator responsiveness test.<br>*If FEV1 (or PEF*) is <70% predicted*: consider starting or stepping up maintenance ICS-containing treatment for 3 months, then reassess symptoms and lung function. If no response, resume previous ICS dose and refer patient for diagnosis and investigation. |
| Few respiratory symptoms, normal lung function, and no variableairflow limitation | **Consider repeating BD responsiveness test** again after withholding bronchodilator asabove or during symptoms. If normal, consider investigation for alternative diagnoses.<br>**Consider stepping down ICS-containing treatment**<br>• *If symptoms emerge and lung function falls*: asthma is confirmed. Step up ICS-containing treatment to previous lowest effective dose.<br>• *If no change in symptoms or lung function at lowest controller step*: consider inceasing maintenance ICS-containing treatment or switching to as-needed-only ICS-formoterol,and monitor patient closely for at least 12 months. |
| Persistent shortness of breath and persistent airflow limitation | **Consider stepping up ICS-containing treatment for 3 months,** then reassess symptoms and lung function. If no response, resume previous ICS dose andrefer patient for further investigation and management, or manage as for patients with features of both asthma and COPD. |

**Box 3 How to step-down ICS-containing treatment to help confirm the diagnosis of asthma**

| 1. ASSESS |
|---|
| • Document the patient's current status including asthma symptom control and risk factors and lung function. If the patient has risk factors for asthma exacerbations, step down treatment only with close supervision.<br>• Choose a suitable time (e.g., no respiratory infection, not going away on vacation, not pregnant).<br>• Provide a written asthma action plan, so the patient/caregiver knows how to recognize and respond if symptoms worsen. Ensure they will have enough medication to be able to resume their previous dose if their asthma worsens after stepping down. |
| **2. ADJUST** |
| • Show the patient/caregiver how to reduce their ICS dose by 25–50%, or stop other maintenance medication (e.g., LABA) if being used. Schedule a review visit for 2–4 weeks. |
| **3. REVIEW RESPONSE** |
| • Repeat assessment of asthma control and lung function tests in 2–4 weeks.<br>• If symptoms increase and variable expiratory airflow limitation is confirmed after stepping down treatment, the diagnosis of asthma is confirmed. The patient should be returned to their lowest previous effective treatment.<br>• If, after stepping down to a low-dose ICS-containing treatment, symptoms do not worsen and there is still no evidence of variable expiratory airflow limitation to confirm the diagnosis of asthma, consider ceasing ICS-containing treatment and repeating asthma control assessment and lung function tests in 2–3 weeks but follow the patient for at least 12 months. |

## Other tests

### Allergy tests

The presence of atopy increases the probability that a patient with respiratory symptoms has allergic asthma, but this is not specific for asthma nor is it present in all asthma phenotypes. Atopic status can be identified by skin prick testing or by measuring the level of specific immunoglobulin E (sIgE) in serum.

### Imaging

Chest X-ray is not routinely recommended unless the diagnosis is in doubt, when symptoms are not typical or suggest alternative diagnoses, assessing patient with severe asthma, or when asthma associated with other medical conditions.

Imaging may also be used to identify congenital abnormalities in infants with asthma-like symptoms, and alternative diagnoses in children with difficult-to-treat asthma. High-resolution computed tomography (CT) of the lungs can identify conditions such as bronchiectasis, emphysema, lung nodules, airway wall thickening and lung distension.

## How to make the diagnosis of asthma in other contexts

### Patients presenting with persistent cough as the only respiratory symptom

Common causes of an isolated non-productive cough include cough-variant asthma, chronic upper airway cough syndrome (often called 'postnasal drip'), cough induced by angiotensin-converting enzyme (ACE) inhibitors, gastroesophageal reflux, chronic sinusitis, post-infectious cough, inducible laryngeal obstruction, and eosinophilic bronchitis.

In cough variant asthma, a persistent cough is the only symptom, or, in cough predominant asthma, the most prominent symptom. The cough may be worse at night or with exercise, and in some patients it is productive. Spirometry is usually normal, and the only abnormality in lung function may be airway hyperresponsiveness on bronchial provocation testing.

Cough-variant asthma must also be distinguished from eosinophilic bronchitis in which patients have cough and sputum eosinophilia but normal spirometry and normal airway responsiveness. Treatment of cough variant asthma follows usual recommendations for asthma.

### Occupational asthma and work-exacerbated asthma

Asthma acquired in the workplace is frequently missed. Asthma may be induced or (more commonly) aggravated by exposure to allergens or other sensitizing agents at work, or sometimes from a single, massive exposure. Occupational rhinitis may precede asthma by up to a year and early diagnosis is essential, as persistent exposure is associated with worse outcomes.

Asking patients whether their symptoms improve when they are away from work (weekends or vacations) is an essential screening question. A specialist referral is usually necessary.

### Pregnant women

Pregnant women and women planning a pregnancy should be asked whether they have asthma so that appropriate advice about asthma management and medications can be given If the clinical history is consistent with asthma, and other diagnoses appear unlikely, but the diagnosis of asthma is not confirmed on initial bronchodilator responsiveness testing, manage as asthma with ICS-containing treatment and postpone other diagnostic investigations until after delivery. During pregnancy, bronchial provocation testing is contraindicated, and it is not advisable to step down ICS-containing treatment.

### The elderly

Asthma is frequently undiagnosed in the elderly, due to poor perception of airflow limitation; acceptance of dyspnea as being 'normal' in old age, lack of fitness, and reduced physical activity. The presence of multimorbidity also complicates the diagnosis. In a large population-based survey of asthma patients older than 65 years, factors associated with a history of asthma hospitalization included co`-diagnosis of COPD, coronary artery disease, depression, diabetes mellitus, and difficulty accessing medications or clinical care because of cost.

Symptoms of wheezing, breathlessness and cough that are worse on exercise or at night can also be caused by cardiovascular disease or left ventricular failure, which are common in this age group. A careful history and physical examination, combined with an electrocardiogram and chest X-ray, will assist in the diagnosis. Measurement of plasma brain natriuretic polypeptide and assessment of cardiac function with echocardiography may also be helpful. In older people with a history of smoking or biomass fuel exposure, COPD and overlapping asthma and COPD (asthma– COPD overlap) should be considered.

### Smokers and ex-smokers

Asthma and COPD may be difficult to distinguish in clinical practice, particularly in older patients and smokers and ex-smokers, and these conditions may overlap (asthma-COPD overlap). defines COPD on the basis of chronic respiratory symptoms, and environmental exposures such as smoking or inhalation of toxic particles or gases, with confirmation by post-bronchodilator FEV1/FVC <0.7. Clinically important bronchodilator reversibility (FEV1 improvement >12% and >200 mL) is often found in Asthma. Low diffusion capacity is more common in COPD than asthma. The history and pattern of symptoms and past records can help to distinguish these patients from those with long-standing asthma who have developed persistent airflow limitation. Uncertainty in the diagnosis should prompt early referral for specialized investigation and treatment recommendations, as patients with asthma-COPD overlap have worse outcomes than those with asthma or COPD alone.

### Obese patients

While asthma is more common in obese than non-obese people, respiratory symptoms associated with obesity can mimic asthma. In obese patients with dyspnea on exertion, it is important to confirm the diagnosis of asthma with objective measurement of variable expiratory airflow limitation.

## Differential Diagnosis

The differential diagnosis in a patient with suspected asthma varies with age, Any of these alternative diagnoses may also be found together with asthma.

**Box 4. Differential diagnosis of asthma in adults, adolescents and children 6–11 years**

| Age | If the symptoms or signs below are present, consider… | Condition |
|---|---|---|
| 6–11 Years | Sneezing, itching, blocked nose, throat-clearing | Chronic upper airway cough syndrome |
| | Sudden onset of symptoms, unilateral wheeze | Inhaled foreign body |
| | Recurrent infections, productive cough | Bronchiectasis |
| | Recurrent infections, productive cough, sinusitis | Primary ciliary dyskinesia |
| | Cardiac murmurs | Congenital heart disease |
| | Pre-term delivery, symptoms since birth | Bronchopulmonary dysplasia |
| | Excessive cough and mucus production, gastrointestinal symptoms | Cystic fibrosis |
| 12–39 Years | Sneezing, itching, blocked nose, throat-clearing | Chronic upper airway cough syndrome |
| | Dyspnea, inspiratory wheezing (stridor) | Inducible laryngeal obstruction |
| | Dizziness, paresthesia, sighing | Hyperventilation, dysfunctional breathing |
| | Productive cough, recurrent infections | Bronchiectasis |
| | Excessive cough and mucus production | Cystic fibrosis |
| | Cardiac murmurs | Congenital heart disease |
| | Shortness of breath, family history of early emphysema | Alpha1-antitrypsin deficiency |
| | Sudden onset of symptoms | Inhaled foreign body |
| 40+ Years | Dyspnea, inspiratory wheezing (stridor) | Inducible laryngeal obstruction |
| | Dizziness, paresthesia, sighing | Hyperventilation, dysfunctional breathing |
| | Cough, sputum, dyspnea on exertion, smoking or noxious exposure | COPD |
| | Productive cough, recurrent infections | Bronchiectasis |
| | Dyspnea with exertion, nocturnal symptoms, ankle edema | Cardiac failure |
| | Treatment with angiotensin-converting enzyme (ACE) inhibitor | Medication-related cough |
| | Dyspnea with exertion, non-productive cough, finger clubbing | Parenchymal lung disease |
| | Sudden onset of dyspnea, chest pain | Pulmonary embolism |
| | Dyspnea, unresponsive to bronchodilators | Central airway obstruction |
| All ages | Chronic cough, hemoptysis, dyspnea; and/or fatigue, fever, (night) sweats, anorexia, weight loss | Tuberculosis |
| | Prolonged paroxysms of coughing, sometimes stridor | Pertussis |

## Asthma control

The level of asthma control is the extent to which the manifestations of asthma can be observed in the patient or have been reduced or removed by treatment. It is determined by the interaction between the patient's genetic background, underlying disease processes, the treatment that they are taking, environment, and psychosocial factors.

Asthma control has two domains: **symptom control** and **future risk of adverse outcomes.** Both should always be assessed. Lung function is an important part of the assessment of future risk; it should be measured at the start of treatment, after 3–6 months of treatment (to identify the patient's personal best), and periodically thereafter for ongoing risk assessment. Poor symptom control is burdensome to patients and increases the risk of exacerbations, but patients with good symptom control can still have severe exacerbations.

### Assessing Asthma severity

The current definition of asthma severity is based on retrospective assessment, after at least 2–3 months of asthma treatment, from the intensity of treatment required to control symptoms and exacerbations. The retrospective definition of mild asthma as 'easy to treat' is less useful, as patients with few interval symptoms can have exacerbations triggered by external factors such as viral infections or allergen exposure, and the treatment that was historically regarded as the lowest intensity short-acting beta2 agonist (SABA) alone actually increases the risk of exacerbations.

### How to assess a patient's asthma

- Assess symptom control from the frequency of daytime and night-time asthma symptoms, night waking and activity limitation and, for patients using SABA reliever, their frequency of SABA use. One of the commonly used tools for assessing recent symptom control is Asthma Control Test (ACT).
- Also, separately, assess the patient's risk factors for exacerbations, even if their symptom control is good. Risk factors for exacerbations that are independent of symptom control include not only a history of ≥1 exacerbation in the previous year, but also SABA-only treatment (without any ICS), over-use of SABA, socioeconomic problems, poor adherence, incorrect inhaler technique, low forced expiratory volume in 1 second (FEV1), exposures such as smoking, and blood eosinophilia.
- Also assess risk factors for persistent airflow limitation and medication side-effects (including from oral corticosteroids), treatment issues such as inhaler technique and adherence, and comorbidities, and ask the patient /caregiver about their asthma goals and treatment preferences.
- Once the diagnosis of asthma has been made, the main role of lung function testing is in the assessment of future risk. It should be recorded at diagnosis, 3–6 months after starting treatment, and periodically thereafter.

**Box 5 Classification of Asthma Severity (Youths >12 years of age and adults)**

| Components of Severity | | Classification of Asthma Severity (Youths >12 years of age and adults) | | | |
|---|---|---|---|---|---|
| | | Intermittent | Persistent | | |
| | | | Mild | Moderate | Severe |
| Impairment<br><br>Normal FEV1/FVC:<br><br>8-19 yr 85%<br>20 -39 yr 80%<br>40 -59 yr 75%<br>60 -80 yr 70% | Symptoms | ≤2 days/week | >2 days/week but not daily | Daily | Throughout the day |
| | Nighttime awakenings | ≤2x/month | 3-4x/month | > Ix /week but not nightly | Often 7x/week |
| | Short-acting beta2-agonist use for symptom control (not prevention of EIB*) | ≤2 days/week | >2 days/week but not >Ix/day | Daily | Several times per day |
| | Interference with normal activity | None | Minor limitation | Some limitation | Extremely limited |
| | Lung function | • Normal FEV1 between exacerbations<br>• FEV1 >80% predicted<br>• FEV1/FVC normal | • FEV1 >80% predicted<br><br>• FEV1/FVC normal | • FEV1 >60% but <80% predicted<br><br>• FEV1/FVC reduced 5% | • FEV1 <60% predicted<br>• FEV1/FVC reduced >5% |

***EIB: - Exercise induce bronchoconstriction**

**Box 6. Summary of assessment of asthma in adults, adolescents, and children 6–11 years**

**1. Assess asthma control symptom control AND future risk of adverse outcomes**

- Assess symptom control over the last 4 weeks (Numerical 'asthma control' tools)
  Asthma Control Test (ACT): Scores range from 5–25 (higher is better). Scores of 20–25 are classified as well-controlled; 16–19 as not well-controlled; and 5–15 as very poorly controlled asthma. The ACT has four symptom/ reliever questions plus patient self-assessed control. The minimum clinically important difference is 3 points .
- Identify any other risk factors for exacerbations, persistent airflow limitation or side-effects.
- Measure lung function at diagnosis/start of treatment, 3–6 months after starting ICS-containing treatment, then periodically, e.g., at least once every 1–2 years, but more often in at-risk patients and those with severe asthma.

**2. Assess treatment issues**

- Document the patient's current treatment step
- Watch inhaler technique, assess adherence and side-effects.
- Check that the patient has a written asthma action plan.
- Ask about the patient's attitudes and goals for their asthma and medications.

**3. Assess multimorbidity**

- Rhinitis, rhinosinusitis, gastroesophageal reflux, obesity, obstructive sleep apnea, depression and anxiety can contribute to symptoms and poor quality of life, and sometimes to poor asthma control.

**Box 7 Assessment of asthma control at clinical visits in adults, adolescents and children 6–11 years**

**A. Recent asthma symptom control**

Asthma Control Test (ACT)

**B. Risk factors for poor asthma outcomes**

Assess risk factors at diagnosis and periodically, particularly for patients experiencing exacerbations.
Measure FEV1 at start of treatment, after 3–6 months of ICS-containing treatment to record the patient's personal bestlung function, then periodically for ongoing risk assessment.

**a. Risk factors for exacerbations**

**Uncontrolled asthma symptoms:** Having uncontrolled symptoms is an important risk factor for exacerbations.

Factors that increase the risk of exacerbations even if the patient has few asthma symptoms†

***SABA over-use:*** High SABA use (≥3 x 200-dose canisters/year associated with increased risk of exacerbations,
increased mortality particularly if ≥1 canister per month)

***Inadequate ICS***: not prescribed ICS, poor adherence, or incorrect inhaler technique

***Other medical conditions:*** Obesity, chronic rhinosinusitis, GERD, confirmed food allergy, pregnancy

***Exposures:*** Smoking, e-cigarettes, allergen exposure if sensitized, air pollution.

***Psychosocial:*** *Major psychological or socioeconomic problems.*

***Lung function:*** Low FEV1 (especially <60% predicted), high bronchodilator responsiveness

***Type 2 inflammatory markers:*** Higher blood eosinophils, high FeNO (adults with allergic asthma on ICS)

***Exacerbation history:*** Ever intubated or in intensive care unit for asthma; ≥1 severe exacerbation in last year

**b. Risk factors for developing persistent airflow limitation**

***History:*** Preterm birth, low birth weight and greater infant weight gain,chronic mucus hypersecretion

***Medications:*** Lack of ICS treatment in patient with history of severe exacerbation

***Exposures:*** Tobacco smoke, noxious chemicals; occupational or domestic exposures

***Investigation findings:*** Low initial FEV1, sputum or blood eosinophilia

**c. Risk factors for medication side-effects**

***Systemic*** Frequent OCS, long-term, high-dose and/or potent ICS, P450 inhibitors

***Local:*** High-dose or potent ICS, poor inhaler technique

**Box 8 Specific questions for assessment of asthma in children 6–11 years**

| | |
|---|---|
| **Asthma symptom control** | |
| **Day symptoms** | Ask: How often does the child have cough, wheeze, dyspnea or heavy breathing (number of times per week or day)? What triggers the symptoms? How are symptoms managed? |
| **Night symptoms** | Cough, awakenings, tiredness during the day. (If the only symptom is nocturnal cough, consider other diagnoses such as rhinitis or gastroesophageal reflux disease). |
| **Reliever use** | How often is reliever medication used? (check date on inhaler or last prescription) Distinguish between pre-exercise use (sports) and use for relief of symptoms. |
| **Level of activity** | What sports/hobbies/interests does the child have, at school and in their spare time? How does the child's level of activity compare with their peers or siblings? How many days is the child absent from school? Try to get an accurate picture of the child's day from the child without interruption from the parent/caregiver. |
| **Risk factors for adverse outcomes** | |
| **Exacerbations** | Ask: How do viral infections affect a child's asthma? Do symptoms interfere with school or sports? How long do the symptoms last? How many episodes have occurred since their last medical review? Any urgent doctor/emergency department visits? Is there a written action plan? Risk factors for exacerbations include a history of exacerbations, poor symptom control, poor adherence and poverty, and persistent bronchodilator reversibility even if the child has few symptoms. |
| **Lung function** | Check spirogram curves and technique. Main focus is on FEV1 and FEV1/FVC ratio. Plot these values as percent predicted to see trends over time. |
| **Side-effects** | Check the child's height at least yearly, as poorly controlled asthma can affect growth, and growth velocity may be lower in the first 1–2 years of ICS treatment. Ask about frequency and dose of ICS and OCS. |
| **Treatment factors** | |
| **Inhaler technique** | Ask the child to show how they use their inhaler. Compare with a device-specific checklist. |
| **Adherence** | Is there any of the child's prescribed maintenance medication (inhalers and/or tablets) in the home at present? On how many days in a week does the child use it (e.g., 0, 2, 4, 7 days)? Is it easier to remember to use it in the morning or evening? Where is the medication kept, is it in plain view to reduce forgetting? Check date on inhaler. |
| **Goals/concerns** | Does the child or their parent or caregiver have any concerns about their asthma (e.g., fear of medication, side-effects, interference with activity)? What are their goals for treatment? |

| Comorbidities | |
|---|---|
| **Allergic rhinitis** | Itching, sneezing, nasal obstruction? Can the child breathe through their nose? What medications are being taken for nasal symptoms? |
| **Eczema** | Sleep disturbance, topical corticosteroids? |
| **Food allergy** | Is the child allergic to any foods? (Confirmed food allergy is a risk factor for asthma-relateddeath.) |
| **Obesity** | Check age-adjusted BMI. Ask about diet and physical activity. |
| **Other investigations (if needed)** | |
| **2-week diary** | If no clear assessment can be made based on the above questions, ask the child or parent/caregiver to keep a daily diary of asthma symptoms, reliever use and peak expiratoryflow (best of three) for 2 weeks. |
| **Formal exercise challenge** | Provides information about airway hyperresponsiveness and fitness. Onlyperform challenge testing if it is otherwise difficult to assess asthma control. |

**How to distinguish between uncontrolled asthma and severe asthma**

The most common problems that need to be excluded before making a diagnosis of severe asthma are:

- Poor inhaler technique (up to 80% of community patients)
- Poor medication adherence
- Incorrect diagnosis of asthma, with symptoms due to alternative conditions such as inducible laryngeal obstruction, cardiac failure or lack of fitness.
- Multimorbidity such as rhinosinusitis, GERD, obesity and obstructive sleep apnea
- Ongoing exposure to sensitizing or irritant agents in the home or work environment.

## How to interpret changes in lung function in clinical practice

With regular ICS treatment, FEV1 starts to improve within days, and reaches a plateau after around 2 months. The patient's highest FEV1 reading (personal best) should be documented, as this provides a more useful comparison for clinical practice than FEV1 percent predicted. If predicted values are used in children, measure their height at each visit.

Some patients may have a faster than average decrease in lung function and develop persistent (incompletely reversible) airflow limitation. While a short-term (e.g., 3 months) trial of higher dose ICS or ICS-LABA may be appropriate to see if FEV1 can be improved, high doses should not be continued longer than this if there is no response.

The between-visit variability of FEV1 (up to 12% week-to-week or 15% year-to-year in healthy individuals) limits its use in adjusting asthma treatment or identifying accelerated decline in clinical practice. The minimal important difference for improvement and worsening in FEV1 based on patient perception of change has been reported to the long-term goal of asthma management from a clinical perspective is to achieve the best possible outcomes for

the patient, including long-term symptom control and long-term asthma risk minimization. This includes preventing exacerbations, accelerated decline in lung function, and medication adverse effects. minimizing asthma deaths, urgent health care utilization, and the socioeconomic impacts of uncontrolled asthma be about 10%.

## Principles of asthma management in adults, adolescents and children 6–11 yrs.

The goal of asthma management is to achieve the best possible long-term asthma outcomes for the patient:

- Long-term asthma symptom control, which may include:
  - Few/no asthma symptoms
  - No sleep disturbance due to asthma
  - Unimpaired physical activity
- Long-term asthma risk minimization, which may include:
  - No exacerbations
  - Improved or stable personal best lung function
  - No requirement for maintenance systemic corticosteroids
  - No medication side-effects.

The long-term goal of asthma treatment is to achieve the best possible long-term outcomes for the patient for more details about goals of treatment. For every patient, assessment of asthma should include the assessment of asthma control (both symptom control and future risk of adverse outcomes), treatment issues (particularly inhaler technique and adherence), and any comorbidities that could contribute to symptom burden and poor quality of life. Asthma control has two domains: symptom control and risk reduction. asthma management has focused not only on asthma symptom control, but also on personalized management of the patient's modifiable risk factors for exacerbations, other adverse outcomes and multimorbidity, while also considering the patient's preferences and goals. Non-modifiable risk factors, such as a history of past ICU admission, should also be documented. Lung function, particularly FEV1 as a percentage of predicted value, is an important part of the assessment of future risk.

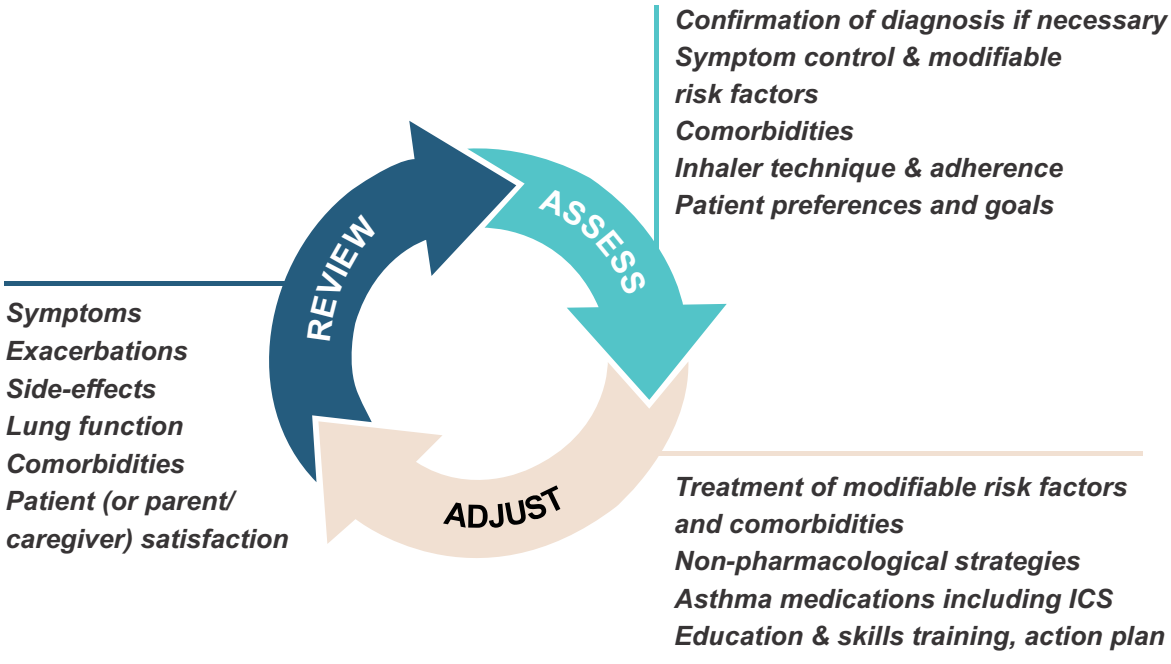
### Personalized control-based asthma management

For many patients in primary care, achieving good symptom control is a good guide to a reduced risk of exacerbations. When ICSs were introduced into asthma management, large improvements were observed in symptom control and lung function, and exacerbations and asthma-related mortality also decreased.

However, ***patients with few or intermittent symptoms may be still at risk of severe exacerbations***, in addition, some patients continue to have exacerbations despite well-controlled symptoms, and for patients with ongoing symptoms, side-effects may be an issue

if ICS doses continue to be stepped up. Therefore, in control- based management, both domains of asthma control (symptom control and future risk; should be considered when choosing asthma treatment and reviewing the response

**Box 9. The asthma management cycle for personalized asthma care**




Personalized asthma management involves a continual cycle of assessment, adjustment of treatment and review

- **ASSESS** the patient's symptom control and their risk factors for exacerbations, for decline in lung function and for medication adverse effects, with particular attention to inhaler technique and adherence. Assess comorbidities and the patient's goals and preferences and confirm the diagnosis of asthma if not yet done.
- **ADJUST** the patient's management, based on these assessments. This includes treatment of modifiable risk factors and comorbidities, relevant non-pharmacologic strategies, education and skills training, and adjustment of medication as required. For adults and adolescents, the preferred controller and reliever treatment across all steps is with combination ICS formoterol, as shown in GINA Track 1.
- **REVIEW** the patient in line with the goals of treatment, reassess factors affecting symptoms, risk of adverse outcomes and patient satisfaction, arrange further investigations if needed, and readjust treatment if needed.

## Minimizing adverse effects of medication

Reduce the potential for local and/or systemic side-effects of inhaled medications by:

- Ensuring correct inhaler technique.
- Reminding patients to rinse and spit out after using ICS.
- Finding each patient's minimum effective dose of ICS-containing therapy (the lowest dose that will, in conjunction with an action plan, maintain good symptom control and minimize exacerbations.
- Checking for drug interactions particularly with cytochrome P450 inhibitors. To reduce the need for OCS, with its multiple cumulative adverse effects, optimize inhaled therapy, including switching treatment to GINA Track 1 with anti-inflammatory reliever therapy (if available). Anti-inflammatory reliever treatment alone (AIR-only) markedly reduces the risk of severe exacerbations requiring OCS compared with SABA alone, and maintenance-and-reliever therapy (MART) with ICS-formoterol reduces the risk of severe exacerbations requiring OCS compared with the same or higher dose of ICS or ICS-LABA or compared with usual care. Treating modifiable risk factors and comorbidities may also reduce the risk of exacerbations and use of OCS.

## Managing other modifiable risk factors

Some patients continue to experience exacerbations even with maximal doses of current treatment. Having even one exacerbation increases the risk that a patient will have another within the next 12 months. There is increasing research interest in identifying at-risk patients, and in investigating new strategies to further reduce exacerbation risk.

In clinical practice, exacerbation risk can be reduced both by optimizing asthma medications, and by identifying and treating modifiable risk factors. Not all risk factors require or respond to a step up in controller treatment.

**Box 10 Treating potentially modifiable risk factors to reduce exacerbations and minimize OCS use**

| Risk factor | Treatment strategy |
|---|---|
| Any patient with one or more risk factors for exacerbations (including poor symptom control) | Ensure patient is prescribed an ICS-containing treatment. |
| | Switch to a regimen with an anti-inflammatory reliever (ICS-formoterol orICS-SABA) if available, as this reduces the risk of severe exacerbations compared with if the reliever is SABA. |
| | Ensure patient has a written action plan appropriate for their health literacy. |
| | Review patient more frequently than low-risk patients. |
| | Check inhaler technique and adherence frequently. |
| | Identify and manage any modifiable risk factors. |
| ≥1 severe exacerbation in last year | Switch to a regimen with an anti-inflammatory reliever (as-needed ICS-formoterol or ICS-SABA) if available, as this reduces the risk of severe exacerbations compared with if the reliever is SABA. |
| | Consider stepping up treatment if no modifiable risk factors. |
| | Identify any avoidable triggers for exacerbations. |
| Exposure to tobacco smoke or e-cigarettes | Encourage smoking cessation by patient/family; provide advice and resources. |
| | Consider higher dose of ICS if asthma poorly controlled. |
| Low FEV1, especially if <60% predicted | Address problems with adherence and inhaler technique |
| | Consider trial of 3 months' treatment with high-dose ICS. |
| | Exclude other lung disease, e.g., COPD. |
| | Refer for expert advice if no improvement. |
| Obesity | Provide strategies for weight reduction |
| | Distinguish asthma symptoms from symptoms due to deconditioning,mechanical restriction, and/or sleep apnea. |
| Confirmed food allergy | Appropriate food avoidance; anaphylaxis action plan; injectable epinephrine; refer for expert advice. |
| Occupational or domestic exposure to Irritants | Remove from exposure as soon as possible. |
| | Refer for expert advice as soon as possible. |
| Sputum eosinophilia despite medium/high ICS (few centers) | Increase ICS dose, independent of level of symptom control. |

## Non-pharmacological strategies

In addition to pharmacological treatments, other strategies should be considered where relevant, to assist in improving symptom control and/or reducing future risk.

**Box 11 Non-pharmacological interventions – summary**

| Intervention | Advice/recommendation |
|---|---|
| Cessation of smoking, environmental tobacco exposure (ETS) and vaping | •At every visit, strongly encourage people with asthma who smoke or vape to quit. Provide access to counseling and smoking cessation programs. |
| | •Advise parents/caregivers of children with asthma not to smoke or vape, and not to allow smoking or vaping in rooms or cars that their children use. |
| | •Strongly encourage people with asthma to avoid environmental smoke exposure. |
| | •Assess smokers/ex-smokers for COPD or overlapping features of asthma and COPD (asthma+COPD, as additional treatment strategies may be required. |
| Physical activity | • Encourage people with asthma to engage in regular physical activity for general health benefits. |
| | •Provide advice about prevention of exercise-induced bronchoconstriction with low-dose ICS-formoterol used as needed and before exercise, or with regular daily ICS |
| | •Provide advice about prevention of breakthrough exercise-induced bronchoconstriction with:<br>• warm-up before exercise<br>• SABA (or ICS-SABA) before exercise<br>• low-dose ICS-formoterol before exercise. |
| | • Regular physical activity improves cardiopulmonary fitness, and can have a small benefit for asthma control and lung function, including swimming in young people with asthma. |
| | • Physical activity interventions in adults with moderate/severe asthma is associated with improved symptoms and quality of life. |
| Pulmonary rehabilitation programs | • Structured outpatient pulmonary rehabilitation programs can improve functional exercise capacity (6-minute walk) and quality of life. |
| Avoidance of occupational or domestic exposures to allergens or irritants | • Ask all patients with adult-onset asthma about their work history and other exposures to irritant gases or particles, including at home. |
| | • In management of occupational asthma, identify and eliminate occupational sensitizers as soon as possible, and remove sensitized patients from any further exposure to these agents. |
| | • Patients with suspected or confirmed occupational asthma should be referred promptly for expert assessment and advice, if available. |

**Box 11(continued) Non-pharmacological interventions – summary**

| Intervention | Advice/recommendation |
|---|---|
| Avoidance of medications that may make asthma worse | • Always ask about asthma before prescribing NSAIDs and advise patients to stop using them if asthma worsens. |
| | • Always ask people with asthma about concomitant medications. |
| | • Aspirin and NSAIDs (non-steroidal anti-inflammatory drugs) are not generally contraindicated unless there is a history of previous reactions to these agents. |
| | • Decide about prescription of oral or ophthalmic beta-blockers on a case-by-case basis. Initiate treatment under close medical supervision by a specialist. |
| | • If cardio selective beta-blockers are indicated for acute coronary events, asthma is not an absolute contra-indication, but the relative risks/benefits should be considered. |
| Healthy diet | • Encourage patients with asthma to consume a diet high in fruit and vegetables for its general health benefits. |
| Avoidance of indoor allergens | • Allergen avoidance is not recommended as a general strategy in asthma. |
| | • For sensitized patients, there is limited evidence of clinical benefit for asthma in most circumstances with single-strategy indoor allergen avoidance. |
| | • Remediation of dampness or mold in homes reduces asthma symptoms and medication use in adults. |
| | • For patients sensitized to house dust mite and/or pets, there is limited evidence of clinical benefit for asthma with avoidance strategies (only in children). |
| | • Allergen avoidance strategies are often complicated and expensive, and there are no validated methods for identifying those who are likely to benefit. |
| Weight reduction | • Include weight reduction in the treatment plan for obese patients with asthma. |
| | • For obese adults with asthma a weight reduction program plus twice-weekly aerobic and strength exercises is more effective for symptom control than weight reduction alone. |
| | • The greatest improvement in asthma outcomes with weight reduction is seen with bariatric surgery. |
| Breathing exercises | • Breathing exercises may be a useful supplement to asthma pharmacotherapy for symptoms and quality of life, but they do not reduce exacerbation risk or have consistent effects on lung function. |
| Avoidance of indoor air pollution | • Encourage people with asthma to use non-polluting heating and cooking sources, and for sources of pollutants to be vented outdoors where possible. |
| Avoidance of outdoor allergens | • For sensitized patients, when pollen and mold counts are highest, closing windows and doors, remaining indoors, and using air conditioning may reduce exposure to outdoor allergens. |

## Referral for expert advice

For most patients' asthma can usually be managed in primary care, but some clinical situations warrant referral for expert advice regarding diagnosis and/or management.

**Box 12 Indications for considering referral for expert advice, where available**

| |
|---|
| **Difficulty confirming the diagnosis of asthma** |
| • Patient has symptoms of chronic infection, or features suggesting a cardiac or other non-pulmonary cause(immediate referral recommended).<br>• Diagnosis is unclear, even after a trial of therapy with ICS or systemic corticosteroids.<br>• Patient has features of both asthma and COPD, and there is doubt about priorities for treatment. |
| **Suspected occupational asthma** |
| • Refer for confirmatory testing and identification of sensitizing or irritant agent, and specific advice abouteliminating exposure and pharmacological treatment. |
| **Persistent or severely uncontrolled asthma or frequent exacerbations** |
| • Symptoms remain uncontrolled, or patient has ongoing exacerbations or low lung function despite correct inhaler technique and good adherence with Step 4 treatment (medium-dose ICS-LABA). Before referral, depending on the clinical context, identify and treat modifiable risk factors and comorbidities.<br>• Patient frequently uses asthma-related health care, e.g., multiple ED visits or urgent primary care visits. |
| **Any risk factors for asthma-related death** |
| • Near-fatal asthma attack (ICU admission, or mechanical ventilation for asthma) at any time in the past.<br>• Suspected or confirmed anaphylaxis or food allergy in a patient with asthma |
| **Evidence of, or risk of, significant treatment side-effects** |
| • Significant side-effects from treatment<br>• Need for long-term oral corticosteroid use<br>• Frequent courses of oral corticosteroids (e.g., two or more courses a year) |
| **Symptoms suggesting complications or sub-types of asthma** |
| • e.g., aspirin-exacerbated respiratory disease; allergic bronchopulmonary aspergillosis (ABPA) |
| **Additional reasons for referral in children 6–11 years** |
| • Doubts about diagnosis of asthma e.g., respiratory symptoms are not responding well to treatment in a childwho was born prematurely<br>• Symptoms or exacerbations that remain uncontrolled despite medium-dose ICS with correct inhaler technique and good adherence<br>• Suspected side-effects of treatment (e.g., growth delay)<br>• Concerns about the child's welfare or well-being |

## Medications and strategies for adults, adolescents and children 6–11 years

### Categories of asthma medications

The pharmacological options for long-term treatment of asthma fall into the following main categories.

**Box 13 Terminology for asthma medications**

| Term | Definition | Notes |
|---|---|---|
| **Maintenance treatment** | Asthma treatment that isprescribed for use everyday (or on a regularly scheduled basis) | Medications intended to be used continuously, even when the person does not have asthma symptoms. Examples include ICS-containing medications (ICS, ICS-LABA, ICS-LABA-LAMA), as well as LTRA† and biologic therapy.<br>The term 'maintenance' describes the prescribed frequency of administration, not a particular class of asthma medicine. |
| **Controller** | Medication targeting both domains of asthma control (symptom control and future risk) | In the past, 'controller' was largely used for ICS-containing medications prescribed for regular daily treatment, so 'controller' and 'maintenance' became almost synonymous. However, this became confusing after the introduction of combination ICS-containing relievers for as-needed use.<br>To avoid confusion, 'ICS-containing treatment' and 'maintenance treatment' have been substituted as appropriate where the intended meaning was unclear. |
| **Reliever** | Asthma inhaler taken as needed, for quick relief of asthma symptoms | Sometimes called rescue inhalers. As well as being used for symptom relief, reliever inhalers can also be used before exercise, to prevent exercise-induced asthma symptoms.<br>Includes SABAs (e.g., salbutamol [albuterol], terbutaline, ICS- salbutamol), as-needed ICS-formoterol, and as-needed ICS-SABA.<br>SABA-containing relievers should not be used for regular maintenance use, or to be taken when the person does not haveasthma symptoms (except before exercise). |

| | | |
|---|---|---|
| **Anti-inflammatory reliever (AIR)** | Reliever inhaler that contains both a low-dose ICS and a rapid-acting bronchodilator | Includes budesonide-formoterol, beclomethasone-formoterol and ICS-salbutamol combinations. Patients can also use AIRs as needed before exercise or allergen exposure to prevent asthma symptoms and bronchoconstriction. Non-formoterol LABAs in combination with ICS cannot be used as relievers. ICS-formoterol should not be used as the reliever with maintenance ICS-non-formoterol LABAs.<br>Some anti-inflammatory relievers can be used as-needed at Steps 1–2 as the person's sole asthma treatment, without a maintenance treatment ('**AIR-only**' treatment). Almost all evidence for this is with ICS-formoterol. Some ICS-formoterol combinations can be used as both maintenance treatment and reliever treatment at Steps 3–5. |
| **Maintenance-and-reliever therapy (MART)** | Treatment regimen in which the patient uses an ICS-formoterol inhaler every day (maintenance dose), and also uses the same medication as needed for relief of asthma symptoms (reliever doses) | **MART** (**M**aintenance-**A**nd-**R**eliever **T**herapy) can be used only with combination ICS-formoterol inhalers such as budesonide-formoterol and beclomethasone-formoterol. Other ICS-formoterol inhalers can also potentially be used, but combinations of ICS with non-formoterol LABAs, or ICS-SABA, cannot be used for MART. MART is also sometimes called SMART (single-inhaler maintenance-and-reliever therapy); the meaning is the same. |

## Adults and adolescents: Asthma treatment tracks

**Box 14 Asthma treatment tracks for adults and adolescents**

### Asthma treatment for adults and adolescents is in two Tracks

In Track 1 (preferred), the reliever is as needed low-dose ICS formoterol, and in Track 2, as-needed SABA or as-needed ICS-SABA.

The reasons for showing treatment in two tracks are:

- to show clinicians how treatment can be stepped up and down using the same reliever at each step.
- because ICS-formoterol cannot be used as the reliever in patients prescribed a combination ICS with non-formoterol LABA, due to lack of evidence about efficacy and safety.

### Track 1: The reliever is as-needed low-dose ICS-formoterol

This is the preferred approach for adults and adolescents, because using low-dose ICS formoterol (an anti-inflammatory reliever; AIR) reduces the risk of severe exacerbations compared with regimens that use SABA as reliever, with similar symptom control. In addition, the treatment regimen is simpler, with patients using a single medication for reliever and for maintenance treatment if prescribed, across treatment steps.

- With this approach, when a patient at any treatment step has asthma symptoms, they use low-dose ICS-formoterol in a single inhaler for symptom relief. In Steps 1–2, this provides their anti-inflammatory therapy.
- In Steps 3–5, patients also take ICS formoterol as their daily maintenance treatment; together, this is called 'maintenance-and-reliever therapy' (MART).

### Track 2: The reliever is as-needed SABA or as-needed ICS-SABA

This is an alternative approach if Track 1 is not possible, or if a patient's asthma is stable with good adherence and no exacerbations on their current therapy. However, before prescribing a regimen with SABA reliever, consider whether the patient is likely to be adherent with their maintenance therapy, as otherwise they will be at higher risk of exacerbations.

- In Step 1, the patient takes a SABA and a low-dose ICS together for symptom relief when symptoms occur (in a combination inhaler, or with the ICS taken immediately after the SABA).
- In Steps 2–5, a SABA (alone) or combination ICS-SABA is used for symptom relief, and the patient takes maintenance ICS-containing medication regularly every day. If the reliever and maintenance medication are in different devices, make sure that the patient can use each inhaler correctly.

### Stepping up and down

Treatment can be stepped up or down along one track, using the same reliever at each step, or it can be switched between tracks, according to the individual patient's needs and preferences. Before stepping up, check for common problems such as incorrect inhaler technique, poor adherence, and environmental exposures, and confirm that the symptoms are due to asthma.

## Initial asthma treatment for adults and adolescents

**Box 15 Initial asthma treatment for adults and adolescents with a diagnosis of asthma**

| Presenting symptoms | Preferred INITIAL treatment (Track 1) | Alternative INITIAL treatment (Track 2) |
|---|---|---|
| Infrequent asthma symptoms,e.g., 1–2 days/week or less | **As-needed low-dose ICS-formoterol** | **Low-dose ICS taken whenever SABA is taken**, in combination or separate inhalers. Such patients are highly unlikely to be adherent with daily ICS. |
| Asthma symptoms less than 3–5 days/week, with normal or mildly reduced lung function | | **Low-dose ICS plus as-needed SABA.** Before choosing this option, consider likely adherence with daily ICS. |
| Asthma symptoms most days (e.g., 4–5 days/week or more); or waking due to asthma once aweek or more, or low lung function. | **Low-dose ICS-formoterol maintenance-and-reliever therapy** (MART) | **Low-dose ICS-LABA plus as-needed SABA** or plus as-neededICS-SABA, **OR** Medium-dose ICS plus as-needed SABA or plus as-needed ICS-SABA. Consider likely adherence withdaily maintenance treatment. |
| Daily asthma symptoms, waking at night with asthma once a week or more, with low lung function | **Medium-dose ICS-formoterol maintenance-and-reliever therapy** (MART) | **Medium- or high-dose ICS-LABA** plus, as-needed SABA or plus as-needed ICS-SABA. Consider likely adherence with daily maintenance treatment. High-dose ICS plus as-needed SABA is another option but adherence isworse than with combination ICS-LABA. |
| Initial asthma presentation isduring an acute exacerbation | Treat as for exacerbation including short course of OCS if severe; commence medium-dose MART | Treat as for exacerbation, including short course of OCSif severe; commence medium- or high-dose ICS-LABA plus as-needed SABA. |

**Before starting initial controller treatment**

- Record evidence for the diagnosis of asthma.
- Record the patient's level of symptom control and risk factors, including lung function
- Consider factors influencing choice between available treatment options, including likely adherence with daily ICS-containing treatment, particularly if the reliever is SABA.
- Choose a suitable inhaler and ensure that the patient can use the inhaler correctly.
- Schedule an appointment for a follow-up visit.

**After starting initial controller treatment**

- Review patient's response after 2–3 months, or earlier depending on clinical urgency.
- See for recommendations for ongoing treatment and other key management issues.
- Check adherence and inhaler technique frequently.
- Step down treatment once good control has been maintained for 3 months.

## Non-recommended bronchodilators

- Anticholinergic agents in the absence of ICS: In adults, inhaled short-acting muscarinic antagonists (SAMA) like ipratropium are potential alternatives to SABA for routine relief of asthma symptoms; however, these agents have a slower onset of action than inhaled SABA. Like SABAs they should not be used without ICS. Use of long-acting muscarinic antagonists (LAMA) in asthma without concomitant ICS is associated with an increased risk of severe exacerbations.

- Formoterol without ICS: The rapid-onset LABA, formoterol, is as effective as SABA as a reliever medication in adults and children, and reduces the risk of severe exacerbations by 15–45%, compared with as-needed SABA in short term, but use of regular or frequent LABA without ICS is strongly discouraged because of the risk of future exacerbations
- **SABA-only treatment is not recommended** for adults, adolescents or children 6–11 years with asthma. Although inhaled SABAs are highly effective for the quick relief of asthma symptoms, patients whose asthma is treated with SABA alone (compared with ICS) are **at increased risk of asthma-related death and of urgent asthma-related healthcare even** if they have good symptom control. The risk of severe exacerbations requiring urgent health care is substantially reduced in adults and adolescents by either as-needed ICS-formoterol, or by regular low-dose ICS with as-needed SABA. The risk of asthma exacerbations and mortality increases incrementally with higher SABA use, including in patients treated with SABA alone.
- **Treatment with oral bronchodilators (e.g. salbutamol tablets or syrups; oral theophylline) is not recommended** for treatment of asthma in any age.

## About asthma treatment for children 6–11 years

**Box 12 Initial asthma treatment for children aged 6–11 years with a diagnosis of asthma.**

| Presenting symptoms | Preferred INITIAL treatment |
|---|---|
| Infrequent asthma symptoms, e.g., 1–2 days/week or less | Low-dose ICS taken whenever SABA is taken in combination or in separate inhalers. |
| Asthma symptoms 2–5 days/week | Low-dose ICS plus as-needed SABA Other options include taking ICS whenever SABA is taken in combination or separate inhalers, or daily LTRA (Evidence A for less effectiveness for exacerbations than ICS). Consider likely adherence with maintenance treatment if reliever is SABA. |
| Asthma symptoms most days (e.g., 4–5days/week); or waking due to asthma once a week or more | Low-dose ICS-LABA plus as needed SABA, OR Medium-dose ICS plus as-needed SABA,OR low-maintenance-and-reliever(MART).<br>Other options include daily low-dose ICS and LTRA, plus as-needed SABA. |
| Daily asthma symptoms, waking at night once or more a week, and lowlung function | Medium-dose ICS-LABA plus as-needed SABA, OR low-dose ICS-formoterol maintenance-and-reliever (MART). |
| Initial asthma presentation is during anacute exacerbation. | Treat as for exacerbation, including a short course of OCS if the exacerbation is severe; commence Step 3 or Step 4 treatment, and arrange follow-up. |
| **Before starting initial controller treatment** | |
| • Record evidence for the diagnosis of asthma, if possible.<br>• Record the child's level of symptom control and risk factors, including lung function.<br>• Consider factors influencing choice between available treatment options.<br>• Choose a suitable inhaler and ensure that the child can use the inhaler correctly.<br>• Schedule an appointment for a follow-up visit. | |
| **After starting initial controller treatment** | |
| • Review child's response after 2–3 months, or earlier depending on clinical urgency.<br>• Step down treatment once good control has been maintained for 3 months . | |

## Not recommended

SABA-only treatment is not recommended in children 6–11 years, as for adults and adolescents.

Oral SABA and theophylline are not recommended because of the higher risk of side-effects and lower efficacy.

## Allergen immunotherapy

Allergen-specific immunotherapy may be considered as add-on therapy for adults and children with asthma who have clinically significant sensitization to aeroallergens, including in those with allergic rhinitis.

## Vaccinations

- Influenza causes significant morbidity and mortality in the general population and contributes to some acute asthma exacerbations. influenza vaccination reduced the risk of asthma exacerbations Advise patients with moderate to severe asthma to receive an influenza vaccination every year, or at least whenvaccination of the general population is advised. Follow local immunization schedules.
- Encourage children, adults and the elderly with asthma to follow their local immunization schedule, including for pneumococcal, pertussis, influenza, RSV and COVID-19 vaccinations..
- COVID-19 vaccination and influenza vaccination may be given on the same day.

## Guided asthma self-management education and skills training

As with other chronic diseases, people with asthma need education and skills training to manage it well. This is most effectively achieved through a partnership between the patient/carer and their healthcare providers. The essential components for this include:

- Choosing the most appropriate inhaler for the patient's asthma treatment: consider available devices, cost, the ability of the patient to use the inhaler after training, environmental impact, and patient satisfaction
- Skills training to use inhaler devices effectively
- Encouraging adherence with medications, appointments and other advice, within an agreed management strategy
- Asthma information
- Training in guided self-management, with self-monitoring of symptoms or peak expiratory flow (PEF), a written asthma action plan to show how to recognize and respond to worsening asthma, and regular review by a healthcare provider or trained healthcare worker.

### Self-monitoring of symptoms and/or peak expiratory flow (PEF)

Patients/careers should be trained to keep track of symptoms (with or without a diary), and notice and take action, if necessary, when symptoms start to worsen. PEF monitoring may sometimes be useful:

➢ **In short-term monitoring**

- After an exacerbation, to monitor recovery
- After a change in treatment, to help in assessing whether the patient has responded
- When symptoms appear excessive (for objective evidence of degree of lung function impairment)
- To assist in identification of occupational or domestic triggers for worsening asthma control

➢ **In long-term monitoring**

- For earlier detection of exacerbations.
- For patients with a history of sudden severe exacerbations.
- For patients who have difficult-to-control or severe asthma.

## Written asthma action plans

Personal written asthma action plans show patients how to make short-term changes to their treatment in response to changes in their symptoms and/or PEF. They also describe how and when to access medical care. 'Written' action plans include printed, digital or pictorial plans (i.e., the patient is given a record of the instructions).

Every asthma patient need to have an action plan, and this should be reviewed and updated by doctor every 3 months or according to physician's opinion.

## The goal of a written asthma action plans is to control the disease, so you can:

- Prevent symptoms.
- Reduce need for medicines.
- Maintain normal activity.
- Sleep through the night.
- Prevent emergencies

## The asthma action plan allows the patient to understand asthma medications:

- Different medications have different functions.
- Use quick relief medicines to quickly relieve the symptoms when they occur.
- Use long term control medicines daily to reduce airway inflammation and prevent future symptoms.

## The asthma action plan also includes:

- Asthma triggers to avoid as much as possible.
- The patient records his/her symptoms, this allows to know the poorly controlled patients.
- When to call the doctor or seek medical attention.
- When need to seek emergency care, should know the life-threatening features of asthma.

AAPs are particularly helpful for patients with moderate or severe persistent asthma, a history of severe exacerbations, or poorly controlled asthma. While the content of each AAP may vary to some extent, typical plans outline which medications and what actions to take in the following 3 zones:

| Three Zones of asthma action plan (AAP) | |
|---|---|
| "Green zone," | Which includes medications taken every day to achieve and maintain good control |
| "Yellow zone" | Which includes which rescue medications to add when asthma gets worse and when to see their provider for follow-up. |
| "Red zone" | Which details what medications to take and how to seek care in the event of an asthma emergency. |

## Managing multimorbidity

Multimorbidity is a common problem in patients with chronic diseases such as asthma. It is associated with worse quality of life, increased healthcare utilization and increased adverse effects of treatment. Multimorbidity is particularly commonamong those with difficult-to-treat or severe asthma. Active management of comorbidities such as rhinosinusitis, obesity and GERD is important, as these conditions may also contribute to respiratory symptom burden and lead to medication interactions.

## Obesity

### *Clinical features*

Being overweight or obese is a risk factor for childhood asthma and wheeze, particularly in girls. Asthma is more difficult to control in obese patients, this may be due to a different type of airway inflammation, contributory comorbidities such as obstructive sleep apnea and GERD, mechanical factors, or other as-yet undefined factors. In addition, lack of fitness and reduction in lung volume due to abdominal fat may contribute to dyspnea.

### *Management*

As for other patients with asthma, ICS are the mainstay of treatment in obese patients, although their response may be reduced. Weight reduction should be included in the treatment plan for obese patients with asthma. Increased exercise alone appears to be insufficient. Weight loss can improve asthma control, lung function, health status and reduces medication needs in obese patients.

## Gastroesophageal reflux disease (GERD)

### *Clinical features*

GERD can cause symptoms such as heartburn and epigastric or chest pain and is also a common cause of dry cough. Symptoms and/or diagnosis of GERD are more common in people with asthma than in the general population, but this may be in part due to cough being attributed to asthma; in addition, some asthma medications such as beta2 agonists and theophylline cause relaxation of the lower esophageal sphincter. Asymptomatic gastroesophageal reflux is not a likely cause of poorly controlled asthma.

### *Management*

In general, benefits of proton pump inhibitors in asthma appear to be limited to patients with both symptomatic reflux and night-time respiratory symptoms. Other treatment options include motility agents, lifestyle changes and fundoplication. In summary, symptomatic reflux should be treated, but patients with poorly controlled asthma should not be treated with anti-reflux therapy unless they also have symptomatic reflux.

## Anxiety and depression

***Clinical features***

Anxiety symptoms and psychiatric disorders, particularly depressive and anxiety disorders, are more prevalent among people with asthma. Psychiatric comorbidity is also associated with worse asthma symptom control and medication adherence, and worse asthma-related quality of life. Anxious and depressive symptoms have been associated with increased asthma-related exacerbations and emergency visits. Panic attacks may be mistaken for asthma.

***Management***

There have been few good quality pharmacological and non-pharmacological treatment trials for anxiety or depression in patients with asthma, and results are inconsistent.

## Food allergy and anaphylaxis

***Clinical features***

Rarely, food allergy is a trigger for asthma symptoms (<2% of people with asthma). In patients with confirmed food- induced allergic reactions (anaphylaxis), co-existing asthma is a strong risk factor for more severe and even fatal reactions.

***Management***

Patients who have a confirmed food allergy that puts them at risk for anaphylaxis must always have an epinephrine auto-injector available and be trained how to use it.

## Allergic rhinitis

***Clinical features***

Evidence clearly supports a link between diseases of the upper and lower airways. Most patients with asthma, either allergic or non-allergic, have concurrent rhinitis, and 10 – 40% of patients with allergic rhinitis have asthma. Depending on sensitization and exposure, allergic rhinitis may be seasonal (e.g., ragweed or grass pollen), or perennial (e.g., HDM allergens, furred pets in the home), or intermittent (e.g., furred pets at other locations). Rhinitis is defined as irritation and inflammation of the mucous membranes of the nose. Allergic rhinitis may be accompanied by ocular symptoms (conjunctivitis).

***Management***

International evidence-based guidelines recommend intranasal corticosteroids for treatment of allergic rhinitis.

## Chronic rhinosinusitis with and without nasal polyps (CRSwNP and CRSsNP)

Rhinosinusitis is defined as inflammation of the nose and paranasal sinuses characterized by more than two symptoms including nasal blockage/obstruction and/or nasal discharge (anterior/posterior nasal drip). Other symptoms may include facial pain/pressure and/or a reduction or loss of smell. Sinusitis rarely occurs in the absence of rhinitis.

Rhinosinusitis is defined as acute when symptoms last <12 weeks with complete resolution, and chronic when symptoms occur on most days for at least 12 weeks without complete

resolution.

Chronic rhinosinusitis is an inflammatory condition of the paranasal sinuses that encompasses two clinically distinct entities: chronic rhinosinusitis without nasal polyps (CRSsNP) and chronic rhinosinusitis with nasal polyps (CRSwNP).

***Management***

intranasal saline irrigations, intranasal corticosteroids delivered by irrigation, drops (only one small study of each) or sprays, oral antibiotics (small studies with erythromycin), and oral corticosteroids) improved sinonasal-specific quality of life in patients with chronic rhinosinusitis (most commonly with nasal polyps) and comorbid asthma. However, people with chronic rhinosinusitis and asthma may have a lesser response to rhinosinusitis treatments, compared with in people who do not have asthma.

Current evidence supports stepwise treatment to manage chronic rhinosinusitis in people with asthma, beginning with topical nasal saline irrigations and topical nasal steroids as the main treatment. Oral antibiotics can be used as needed after considering the risks and microbial resistance. Oral corticosteroid treatment is effective but should be minimized due to adverse effects. In patients with CRSwNP, omalizumab, mepolizumab and dupilumab improved subjective and objective assessments including nasal symptoms and polyp size, compared with placebo.

## Managing Asthma in specific populations or settings

### Adolescents

***Clinical features***

Care of teenagers with asthma should consider the rapid physical, emotional, cognitive and social changes that occur during adolescence. Asthma control may improve or worsen, although remission of asthma is seen more commonly in males than females. Exploratory and risk-taking behaviors such as smoking occur at a higher rate in adolescents with chronic diseases than in healthy adolescents.

***Management***

Adolescents and their parent/caregivers should be encouraged in the transition towards asthma self-management by the adolescent, increase self-management, focusing consultations on areas in which the young person is not confident.

Medication regimens should be tailored to the adolescent's needs and lifestyle, and reviews arranged regularly so that the medication regimen can be adjusted for changing needs. In adolescents with mild asthma, use of as-needed low-dose ICS formoterol reduced risk of severe exacerbations compared with SABA alone, and without the need for daily treatment. Change in height from baseline in younger adolescents was significantly greater with as-needed ICS-formoterol than with daily low- dose ICS plus as-needed SABA.

### Exercise-induced bronchoconstriction (EIB)

## Exercise-induced bronchoconstriction (EIB)

***Clinical features***

Physical activity is an important stimulus for asthma symptoms for many patients, with symptoms and bronchoconstriction typically worsening after cessation of exercise.However, shortness of breath or wheezing during exercise may also relate to obesity or a lack of fitness, or to comorbid or alternative conditions such as inducible laryngeal obstruction.

***Management***

Regular treatment with ICS significantly reduces EIB. Training and sufficient warm-up reduce the incidence and severity of EIB. Taking SABAs, LABAs or chromones prior to exercise prevents EIB, but tolerance to the protective effects of SABAs and LABAs against EIB develops with regular (more than once-daily) use. Patients with mild asthma who are prescribed as-needed low- dose ICS-formoterol to prevent exacerbations and control symptoms can use the same medication prior to exercise, if needed, and do not need to be prescribed a SABA for pre-exercise use.

## Athletes

***Clinical features***

Athletes, particularly those competing at a high level, have an increased prevalence of various respiratory conditions compared to non-athletes. They experience a higher prevalence of asthma, EIB, allergic or non-allergic rhinitis, chronic cough, inducible laryngeal obstruction, and recurrent respiratory infections. Airway hyperresponsiveness is common in athletes, often without reported symptoms.

***Management***

Preventive measures to avoid high exposure to air pollutants, allergens (if sensitized) and chlorine levels in pools, particularly during training periods, should be discussed with the athlete. They should avoid training in extreme cold or pollution, and the effects of any therapeutic trials of asthma medications should be documented. Adequate anti-inflammatory therapy, especially ICS, is advised; minimization of use of beta2 agonists will help to avoid the development of tolerance.

## Pregnancy

***Clinical features***

Asthma control often changes during pregnancy; in approximately one-third of women asthma symptoms worsen, in one- third they improve, and in the remaining one-third they remain unchanged. Exacerbations are common in pregnancy, particularly in the second trimester. Exacerbations and poor asthma control during pregnancy may be due to mechanical or hormonal changes, or to cessation or reduction of asthma medications due to concerns by the mother and/or the healthcare provider. Pregnant women appear to be particularly susceptible to the effects of viral respiratory infections, including influenza.

Exacerbations and poor symptom control are associated with worse outcomes for both the baby (pre-term delivery, low birth weight, increased perinatal mortality) and the mother (pre-eclampsia). Risk factors for asthma exacerbations during pregnancy include severe asthma,

multiparity, black ethnicity, depression and anxiety, current smoking, age >35 years and obesity. If asthma is well controlled throughout pregnancy, there is little or no increased risk of adverse maternal or fetal complications.

***Management***

Although there is a general concern about any medication use in pregnancy, the advantages of actively treating asthma in pregnancy markedly outweigh any potential risks of usual asthma medications. For this reason, using medications to achieve good symptom control and prevent exacerbations is justified even when their safety in pregnancy has not been unequivocally proven. Use of ICS, beta2 agonists, montelukast or theophylline is not associated with an increased incidence of fetal abnormalities.

Women with asthma who are pregnant or planning pregnancy should be advised not to stop ICS-containing therapy. Importantly, ICS reduces the risk of exacerbations of asthma during pregnancy and cessation of ICS during pregnancy is a significant risk factor for exacerbations.

Pregnant patients with asthma should be advised that poorly controlled asthma, and exacerbations, provide a much greater risk to their baby than do current asthma treatments. During pregnancy, monitoring of asthma every 4–6 weeks is recommended.

## The elderly

***Clinical features***

Lung function generally decreases with longer duration of asthma and increasing age, due to stiffness of the chest wall, reduced respiratory muscle function, loss of elastic recoil and airway wall remodeling. Older patients may not report asthma symptoms and may attribute breathlessness to normal aging or comorbidities such as cardiovascular disease and obesity. Among the elderly, there is no increased risk of cardiovascular disease among those with asthma, compared with those without asthma, except in current or former smokers. Comorbid arthritis may contribute to reduced exercise capacity and lack of fitness and make inhaler device use difficult. Asthma costs may be higher amongst older patients, because of higher hospitalization rates and medication costs.

***Management***

Decisions about management of asthma in older people with asthma need to consider both the usual goals of symptom control and risk minimization and the impact of comorbidities, concurrent treatments and lack of self- management skills. Side-effects of beta2 agonists such as cardiotoxicity, and corticosteroid side- effects such as skin bruising, osteoporosis and fragility fractures, and cataracts, are more common in the elderly than in younger adults. Clearance of theophylline is also reduced. Elderly patients should be asked about all the other medications they are taking, including eye-drops, and potential drug interactions should be considered. Factors such as arthritis, muscle weakness, impaired vision and inspiratory flow should be considered when choosing inhaler devices for older patients, and inhaler techniques should be checked at each visit. Older patients may have difficulties with complex medication regimens, and prescribing of multiple inhaler devices should be avoided if possible. Large-

print versions may be needed for written information such as asthma action plans. Patients with cognitive impairment may require a carer to help them use their asthma medications.

## Aspirin-exacerbated respiratory disease (AERD)

***Clinical features***

The clinical picture and course of AERD (previously called aspirin-induced asthma) are well established. It starts with nasal congestion and anosmia and progresses to chronic rhinosinusitis with nasal polyps that re-grow rapidly after surgery. Asthma and hypersensitivity to aspirin and nonsteroidal anti-inflammatory drugs (NSAIDs) develop subsequently. Following ingestion of aspirin or NSAIDs, an acute asthma attack develops within minutes to 1–2 hours. It is usually accompanied by rhinorrhea, nasal obstruction, conjunctival irritation, and scarlet flush of the head and neck, and may sometimes progress to severe bronchospasm, shock, loss of consciousness, and respiratory arrest. AERD is more likely to be associated with low lung function and severe asthma, and with increased need for emergency care.

***Management***

Patients with AERD should avoid aspirin or NSAID-containing products and other medications that inhibit cyclooxygenase- 1 (COX-1), but this does not prevent progression of the disease. Where an NSAID is indicated for other medical conditions, a COX-2 inhibitor (e.g., celecoxib or etoricoxib), or paracetamol (acetaminophen), may be considered, ICS are the mainstay of asthma therapy in AERD, but OCS are sometimes required; LTRA may also be useful.

## Allergic bronchopulmonary aspergillosis (ABPA)

***Clinical features***

Allergic bronchopulmonary aspergillosis (ABPA) is a complex pulmonary disease characterized by repeated episodes of wheezing, fleeting pulmonary opacities and development of bronchiectasis, sometimes with malaise, weight loss and hemoptysis. Some patients expectorate brownish sputum plugs. ABPA is most commonly diagnosed in people with asthma or cystic fibrosis, due to a hypersensitivity response to Aspergillus fumigatus, a common indoor and outdoor mold.

***Management***

Current first-line therapy is with oral corticosteroids (e.g., a 4-month tapering course), with itraconazole reserved for those with exacerbations or requiring long-term OC. Clinicians should be aware of the potential for drug interactions between itraconazole (a cytochrome P450 inhibitor) and asthma medications. These interactions may lead to increased risk of ICS adverse effects such as adrenal suppression and Cushing's syndrome and may increase the risk of cardiovascular adverse effects of some LABAs (salmeterol and vilanterol).

## Asthma-COPD overlap

Asthma-COPD overlap' and 'asthma +COPD' are terms used to collectively describe patients who have persistent airflow limitation together with clinical features that are consistent with both asthma and COPD. In epidemiological studies, reported prevalence rates for asthma + COPD have ranged between 9% and 55% of those witheither diagnosis, with variation by gender and age.

## Why are the labels 'asthma' and 'COPD' still important?

- There are extremely important differences in evidence-based treatment recommendations for asthma and COPD. Forexample, treatment with a long-acting beta2 agonist (LABA) and/or long-acting muscarinic antagonist (LAMA) alone (i.e., without inhaled corticosteroids [ICS]) is recommended as initial treatment in COPD but contraindicated in asthma due to the risk of severe exacerbations and death.
- These risks are also seen in patients who have diagnoses of both asthma and COPD, making it important to identifyadult patients who, for safety, should not be treated with long-acting bronchodilators alone. ICS reduce mortality and hospitalizations in patients with asthma, including in those with concomitant COPD.

### Many patients have features of both asthma and COPD

- Distinguishing asthma from COPD can be difficult, particularly in smokers and older adults, and some patients mayhave features of both asthma and COPD.
- The terms 'asthma-COPD overlap' or 'asthma+COPD' are simple descriptors for patients who have features of both asthma and COPD.
- These terms do *not* refer to a single disease entity. They include patients with several clinical phenotypes that arelikely caused by a range of different underlying mechanisms.
- More research is needed to better define these phenotypes and mechanisms, but in the meantime, safety ofpharmacologic treatment is a high priority.

### Diagnosis

- Diagnosis in patients with chronic respiratory symptoms involves a stepwise approach, first, is the patient likely to have chronic airways disease, then syndromic categorization as having typical asthma, typical COPD, features of both, and/or with other conditions such as bronchiectasis.
- Lung function testing is essential for confirming persistent airflow limitation, but variable airflow obstruction can be detected with serial peak flow measurements and/or measurements before and after bronchodilator.

## Initial treatment for safety and clinical efficacy

- **For asthma**: ICS are essential, either alone or in combination with a LABA, to reduce the risk of severe exacerbations and death. Do not treat with LABA and/or LAMA alone, without ICS.
- For patients with features of **both asthma and COPD**, treat as asthma. ICS-containing therapy is important to reduce the risk of severe exacerbations and death. Do not give LABA and/or LAMA alone without ICS.
- **For COPD**: Treat according to current recommendations from the Global Initiative for Chronic Obstructive Lung Disease (GOLD), i.e., initial treatment with LAMA and LABA, plus as-needed SABA; add ICS for patients with hospitalizations, ≥2 exacerbations/year requiring oral corticosteroids (OCS), or blood eosinophils ≥300. Avoid high dose ICS because of risk of pneumonia.
- **All patients**: provide structured education especially focusing on inhaler technique and adherence; assess for, andtreat, other clinical problems, including advice about smoking cessation, immunizations, physical activity, and management of multimorbidity.
- **Specialist referral** for additional investigations in patients with asthma+COPD is encouraged, as they often haveworse outcomes than patients with asthma or COPD alone.

**Box 13 Spirometric measures in asthma and COPD**

| Spirometric variable | Asthma | COPD | Asthma + COPD |
|---|---|---|---|
| Normal FEV1/FVC pre- or post BD | Compatible with asthma. If patient is symptomatic at atime when lung function is normal, consider alternative diagnosis. | Not compatible with COPD | Not compatible |
| Reduced post-BD FEV1/FVC(< lower limit of normal, or <0.7) [GOLD 2024] | Indicates airflow limitation but may improve spontaneously or on treatment | Required for diagnosis of COPD | Required for diagnosis of asthma+COPD |
| Post-BD FEV1 ≥80% predicted | Compatible with diagnosis of asthma (good asthma control or interval between symptoms) | Compatible with mild persistent airflow limitation if post-BD FEV1/FVC is reduced | Compatible with mild persistent airflow limitation if post-BD FEV1/FVC is reduced |
| Post-BD FEV1 <80% predicted | Compatible with diagnosis of asthma. Risk factor for asthma exacerbations | An indicator of severity of airflow limitation and risk of future events (e.g., mortalityand COPD exacerbations) | As for COPD and asthma |
| Post-BD increase in FEV1 ≥12% and 200 mL from baseline (reversible airflow limitation). | Usual at some time in course of asthma, but may not be present when well controlled or on ICS-containing therapy | Common and more likelywhen FEV1 is low | Common and more likely when FEV1 is low |
| Post-BD increase in FEV1 >12% and 400 mL from baseline (marked reversibility) | High probability of asthma | Unusual in COPD | Compatible with asthma+ COPD |

## Management of worsening asthma and exacerbations in adults, adolescents and children 6–11 years

### Definition of asthma exacerbations

Exacerbations of asthma are episodes characterized by a progressive increase in symptoms of shortness of breath, cough, wheezing or chest tightness and progressive decrease in lung function, i.e., they represent a change from the patient's usual status that is sufficient to require a change in treatment. Exacerbations may occur in patients with a pre-existing diagnosis of asthma or, occasionally, as the first presentation of asthma.

### What triggers asthma exacerbations?

Exacerbations usually occur in response to exposure to an external agent (e.g., viral upper respiratory tract infection, pollen or pollution) and/or poor adherence with ICS-containing medication; however, a subset of patients present more acutely and without exposure to known risk factors. Severe exacerbations can occur in patients with mild or well- controlled asthma symptoms. lists factors that increase a patient's risk of exacerbations, independentof their level of symptom control.

Common exacerbation triggers include:

- Viral respiratory infections, e.g., rhinovirus, influenza, adenovirus, pertussis, respiratory syncytial virus
- Allergen exposure e.g., grass pollen and other pollens, soybean dust, fungal spores
- Food allergy
- Outdoor air pollution
- Seasonal changes and/or returning to school in fall (autumn)
- Poor adherence with ICS
- Epidemics of severe asthma exacerbations may occur suddenly, putting high pressure on local health system responses. Such epidemics have been reported in association with springtime thunderstorms and either rye grasspollen or fungal spores, and with environmental exposure to soybean dust.

### Written asthma action plans

- All patients should be provided with a written (i.e., printed, digital or pictorial) asthma action plan appropriate for theirage, their current treatment regimen and their reliever inhaler (short-acting beta2 agonist [SABA] or combination inhaled corticosteroids [ICS]-formoterol), their level of asthma control, and their health literacy, so they know how to recognize and respond to worsening asthma.
- On the action plan, state when and how to change reliever and/or maintenance medications, use oral corticosteroids (OCS) if needed, and access medical care if symptoms fail to respond to treatment.

- Advise patients who have a history of rapid deterioration to go to an acute care facility or see their doctor immediatelytheir asthma starts to worsen.
- Base the action plan on changes in symptoms or (only in adults) peak expiratory flow (PEF).

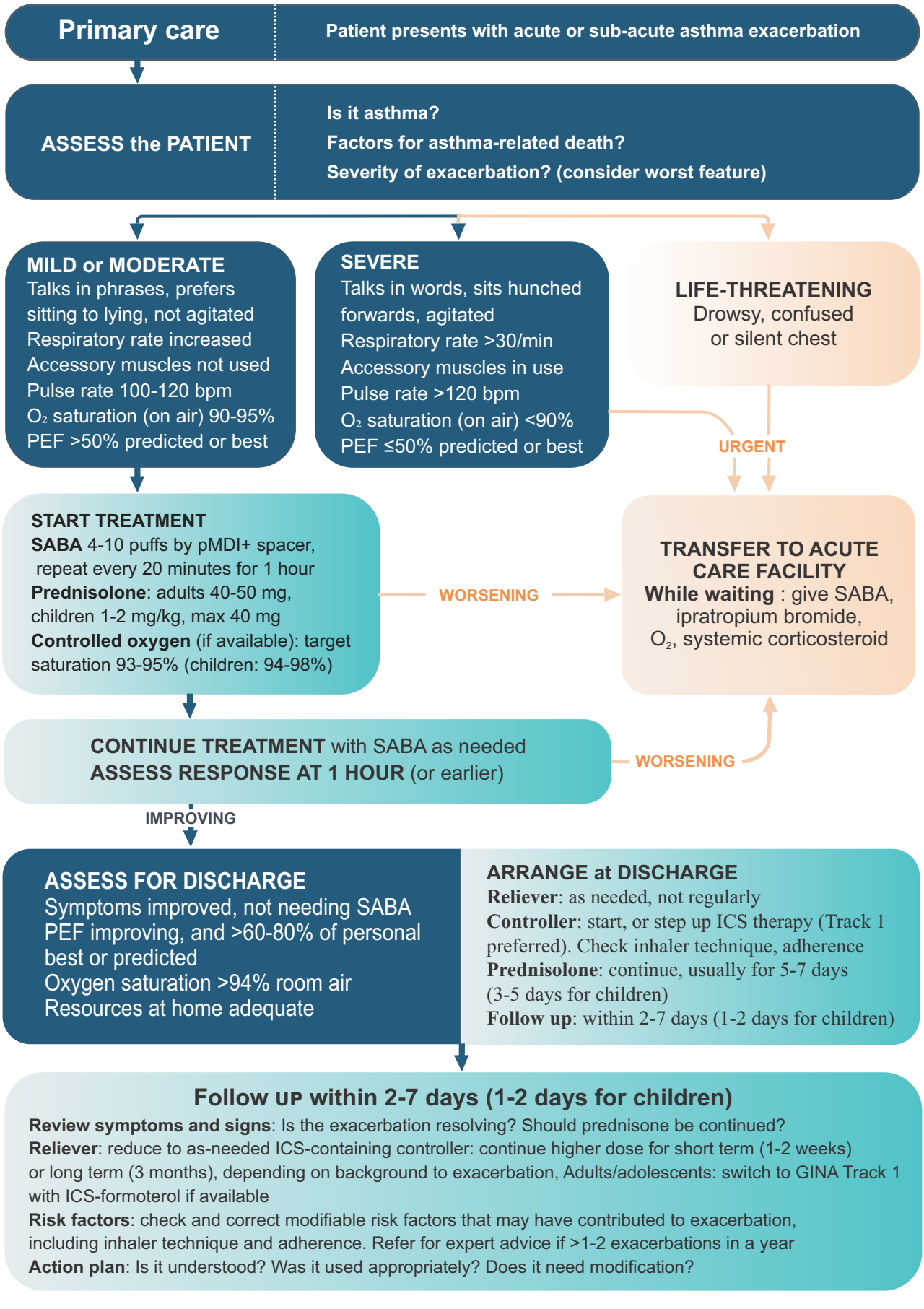
## Identifying patients at risk of asthma-related death

In addition to factors known to increase the risk of asthma exacerbations, some features are specificallyassociated with an increase in the risk of asthma-related death. The presence of one or more of theserisk factors should be quickly identifiable in the clinical notes, and these patients should be encouraged to seek urgent medical care early in the course of an exacerbation.

**Box 14 Factors associated with increased risk of asthma-related death**

- A history of near-fatal asthma requiring intubation and mechanical ventilation.
- Hospitalization or emergency care visit for asthma in the past year.
- Currently using or having recently stopped using oral corticosteroids (a marker of event severity)
- Not currently using inhaled corticosteroids.
- Over-use of short-acting beta2 agonists (SABAs), especially use of an average of more than one canister ofsalbutamol (or equivalent) per month, or using nebulized SABA
- Poor adherence with ICS-containing medications and/or poor adherence with (or lack of) a written asthma actionplan.
- A history of psychiatric disease or psychosocial problems.
- Food allergy in a patient with asthma.
- Several comorbidities including pneumonia, diabetes and arrhythmias were independently associated with anincreased risk of death after hospitalization for an asthma exacerbation.

**Box 15 Management of asthma exacerbations in primary care (adults, adolescents, children 6–11 years)**



## References


1. Global Initiative for Asthma. 2024 GINA Main Report - Global Initiative for Asthma [Internet]. Global Initiative for Asthma. 2024 [cited 2024 Oct 1]. 1–217 p. Available from: https://ginasthma.org/2024-gina-main-report/
2. The Global Asthma Report 2022. Int J Tuberc Lung Dis [Internet]. 2022 [cited 2024 Oct 1];26(1):1–104. Available from: https://globalasthmareport.org/burden/burden.php
3. The Lancet. Global Burden of Disease: GBD cause and risk summaries [Internet]. The Lancet. 2020 [cited 2024 Oct 1]. p. 1129–306. Available from: https://www.thelancet.com/gbd/summaries
4. https://www.pulmonologyadvisor.com/home/decision-support-in-medicine/pulmonary-medicine/asthma-clinical-manifestations-and-management/ Classification of Asthma Severity.
5. The Saudi Initiative for Asthma - 2024 update Guidelines for the Diagnosis and Management of Asthma in Adults and Children.
6. Biological treatments for severe asthma: A major advance in asthma care - Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/Mechanisms-and-characteristic-pathological-features-of-asthma-immunopathology-Features_fig2_331185801

# ANNEXES

## Annex 1: Peak expiratory flow rate - pediatric normal values

| Height (m) | Height (ft) | Predicted **EU** PEFR (L/min) | | Height (m) | Height (ft) | Predicted **EU** PEFR (L/min) |
|---|---|---|---|---|---|---|
| **0.85** | 2'9" | 87 | | **1.30** | 4'3" | 212 |
| **0.90** | 2'11" | 95 | | **1.35** | 4'5" | 233 |
| **0.95** | 3'1" | 104 | | **1.40** | 4'7" | 254 |
| **1.00** | 3'3" | 115 | | **1.45** | 4'9" | 276 |
| **1.05** | 3'5" | 127 | | **1.50** | 4'11 | 299 |
| **1.10** | 3'7" | 141 | | **1.55** | 5'1 | 323 |
| **1.15** | 3'9" | 157 | | **1.60** | 5'3" | 346 |
| **1.20** | 3'11" | 174 | | **1.65** | 5'5" | 370 |
| **1.25** | 4'1" | 192 | | **1.70** | 5'7" | 393 |

### Peak Expiratory Flow Rate:

Measurement of peak flow as a baseline for **diagnosis and monitoring** of asthma

1- Peak flow measurement is effort dependent and requires patient understanding. In certain patients measuring PEF prior to and after a bronchodilator or an exercise challenge may help in confirming the diagnosis. Increase of 20 % after bronchodilator indicates confirmation of asthma diagnosis.

$$\textbf{Percentage of prediction} = \frac{\textbf{PEFR Actual L/min}}{\textbf{PEFR Predicted L/min}} \times 100$$

$$\textbf{Percentage of post bronchodilator improvement} = \frac{\textbf{Post bronchodilator value – Pre bronchodilator value}}{\textbf{Mean of 2 readings}} \times 100$$

2- Measurement of PEF variability by comparing the morning and evening PEF over a period of 2 weeks is also helpful. Variability FOR:

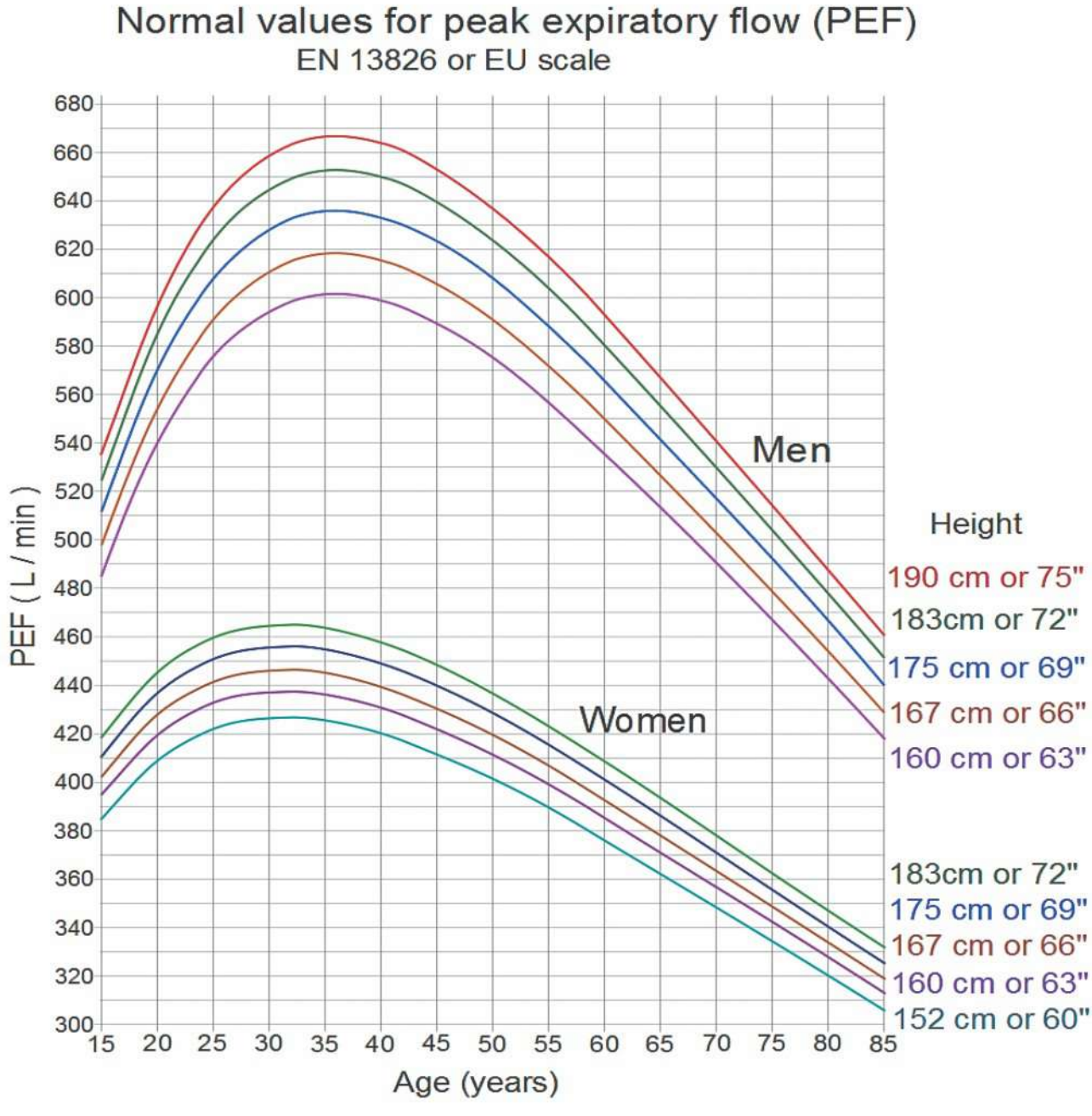
3- Adults: average daily diurnal PEF variability >10%, Children: average daily diurnal PEF variability >13%, variability over 20% make the diagnosis highly probable.

4- It is calculated as follows

$$\textbf{Peak daily flow variability \%} = \frac{\textbf{(highest value – lowest value)}}{\textbf{Mean of 2 readings}}$$

$$\textbf{Diurnal variability over 1-2 weeks} = \frac{\textbf{Sum of readings of daily variability}}{\textbf{No of days}}$$

**Annex 2- Peak expiratory flow rate - normal values (Adults)**

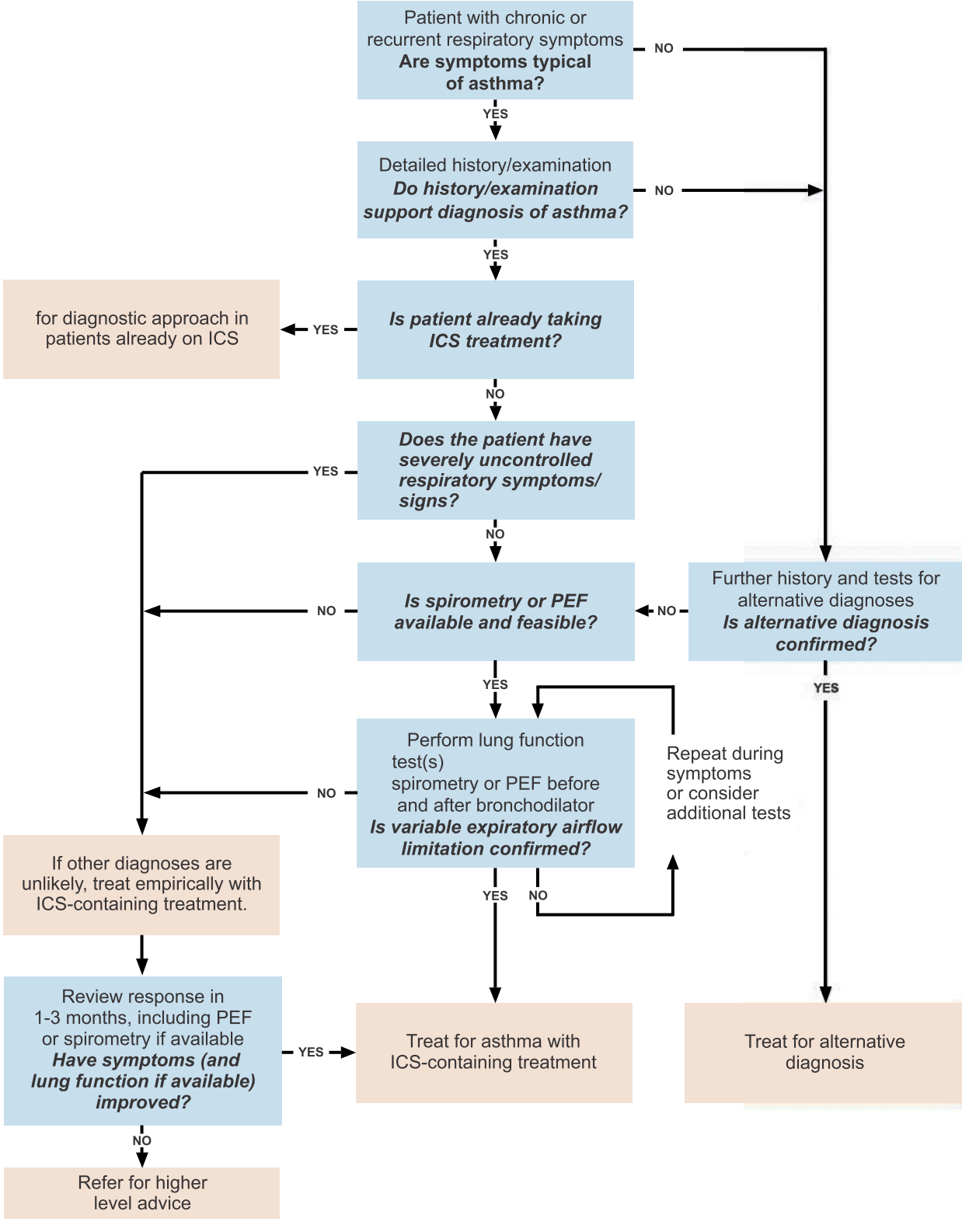


## Annex 3 Asthma Action Plan (AAP)



## Annex 4 Diagnostic flowchart for adults, adolescents and children 6–11 years in clinical practice

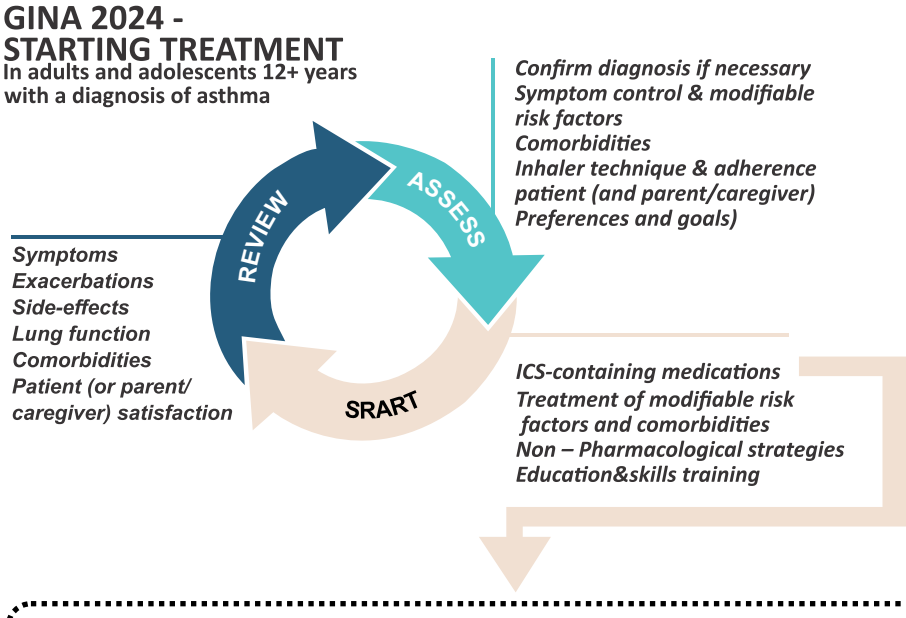
This flowchart is for patients presenting with chronic or recurrent respiratory symptoms in clinical practice.

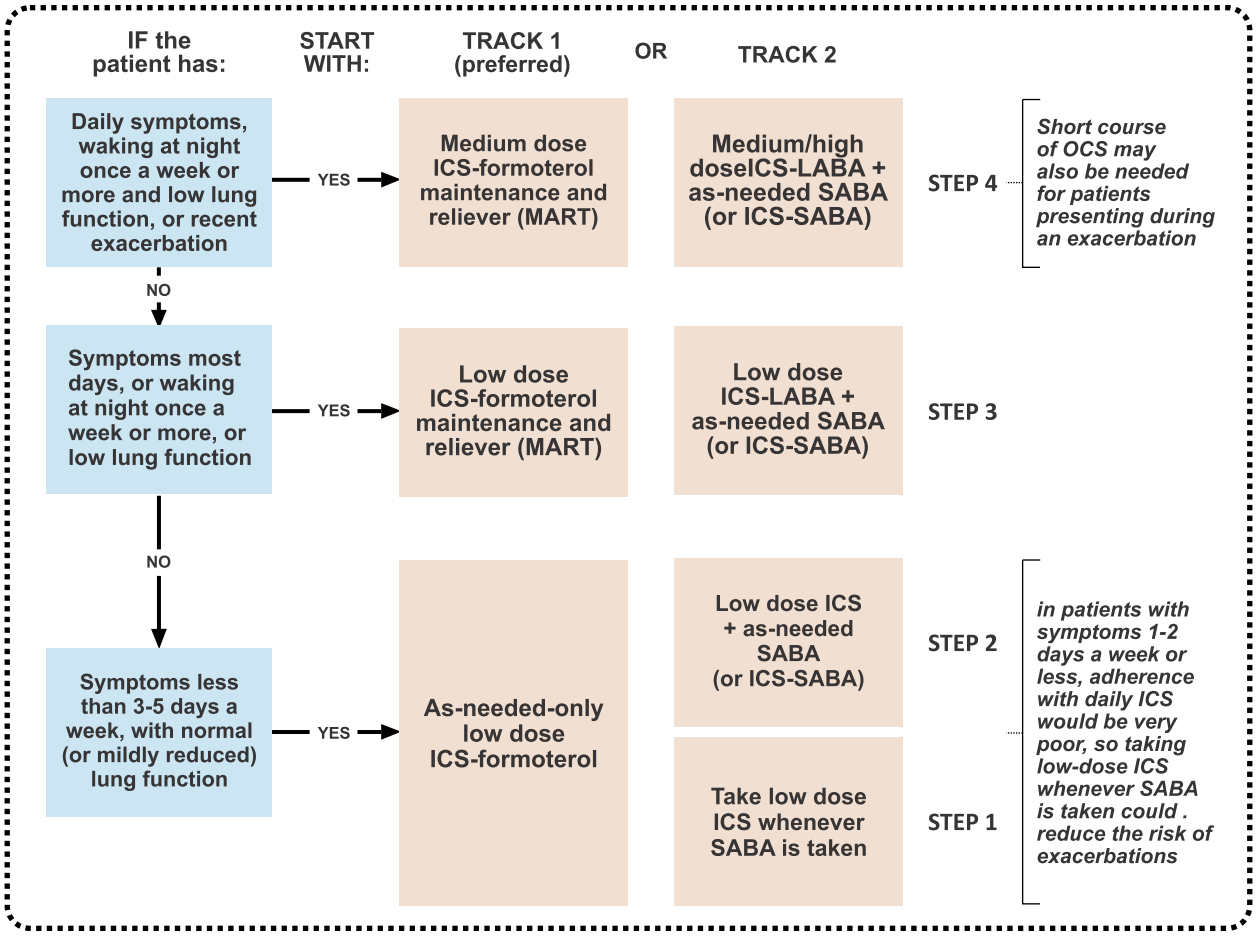


# Annex 5 Investigating poor symptom control and/or exacerbations despite treatment

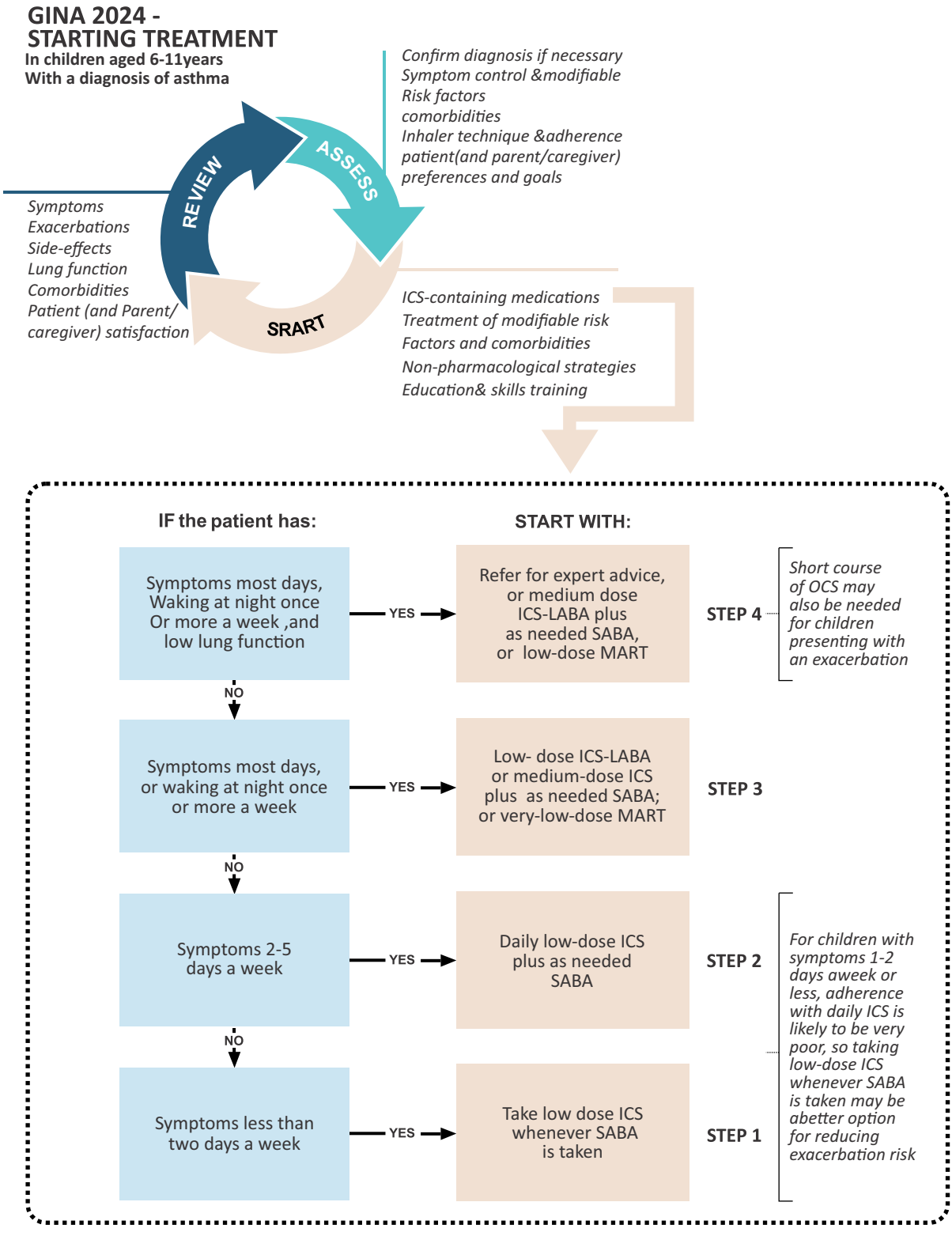
| | |
|---|---|
| **Watch patient using their inhaler Discuss adherence and barriers to use** | • Watch patient use their inhaler(s), check against inhaler checklist. Show correct method, and recheck, up to 3 times. Re-check each visit.<br>• Have empathic discussion to identify poor adherence with maintenance treatment, eg. "Many patients don't use their inhaler as prescribed. In the last 4 weeks, how many days a week have you taken it?" (0 days, 1, 2, 3 etc) and/or: "Do you find it easier to remember your inhaler in the morning or the evening?" Ask about beliefs, cost of medications, and refill frequency |
| **Confirm the diagnosis of asthma** | • If no evidence of variable airflow limitation on spirometry or other testing consider halving ICS dose and repeating lung function after 2-3 weeks check patient has action plan. Consider referring for challenge test |
| **If possible, remove potential risk factors Assess and manage comorbidities** | • For adults/adolescents, switch to GINA Track 1, if available, to reduce exacerbations and simplify regimen<br>• Check for risk factors or inducers such as smoking, bela-blockers or NSAIDs, or occupational or domestic allergen exposure and address as possible<br>• Check for and manage comorbidities (e.g. rhinitis, obesity, GERD, obstructive sleep apnea, depression/anxiety) that may be contrbuting to symptoms or exacerbations |
| **Consider treatment step-up** | • Consider short-term (3-6 months) step-up to next treatment level or alternative option• on present level<br>• Use shared decision-making, and balance potential benefits and risks |
| **Refer for expert advice** | • If asthma still uncontrolled after 3-6 months on high dose ICS-LABA, or with ongoing, risk factors, refer for expert advice<br>• Refer earlier than 6 months if asthma very severe or difficult to manage, or if doubts about diagnosis, or if occupational asthma is suspected |

# Annex 6 Flowchart for selecting initial treatment in adults and adolescents with a diagnosis of asthma



# Annex 7 Flowchart for selecting initial treatment in children aged 6–11 years with a diagnosis of asthma



## Annex 8. Low, medium and high daily metered doses of inhaled corticosteroids (alone or with LABA)

suggested total daily doses for 'low', 'medium' and 'high' dose ICS optionsfor adults/adolescents and children 6–11 years,based on product information.

The table does NOT imply potency equivalence. For example, if you switch treatment from a 'medium' dose of one ICS to a 'medium' dose of another ICS, this may represent a decrease (or an increase) in potency, and the patient's asthma may become unstable (or they may be at increased risk of adverse effects).

Patients should be monitored to ensure stability after any change of treatment or inhaler device. Doses and potency may also differ by country, depending on local products, inhaler devices, regulatory labelling and clinical guidelines or, for one product, with addition of a LAMA to an ICS-LABA.

**Low-dose ICS** provides most of the clinical benefit of ICS for most patients with asthma. However, ICS responsiveness varies between patients, so some patients may need **medium-dose ICS** if their asthma is uncontrolled, or they have ongoing exacerbations, despite good adherence and correct technique with low-dose ICS (with or without LABA). **High-dose ICS** (in combination with LABA or separately) is needed by very few patients, and its long-term use is associated with an increased risk of local and systemic side-effects, which must be balanced against the potential benefits. The timing of medication use also affects outcomes, particularly for exacerbations, as seen with an anti-inflammatory reliever in GINA Track 1.

Daily doses in this table are shown as metered doses. See product information for delivered doses.

| Inhaled corticosteroid (alone or in combination with LABA) | Total daily ICS dose (mcg) – see notes above | | |
|---|---|---|---|
| | Low | Medium | High |
| **Adults and adolescents (12 years and older)** | | | |
| Beclometasone dipropionate (pMDI, standard particle, HFA) | 200–500 | >500–1000 | >1000 |
| Beclometasone dipropionate (DPI or pMDI, extrafine particle, HFA) | 100–200 | >200–400 | >400 |
| Budesonide (DPI, or pMDI, standard particle, HFA) | 200–400 | >400–800 | >800 |
| Ciclesonide (pMDI, extrafine particle, HFA) | 80–160 | >160–320 | >320 |
| Fluticasone furoate (DPI) | 100 | | 200 |
| Fluticasone propionate (DPI) | 100–250 | >250–500 | >500 |
| Fluticasone propionate (pMDI, standard particle, HFA) | 100–250 | >250–500 | >500 |
| Mometasone furoate (DPI) | Depends on DPI device | | |
| Mometasone furoate (pMDI, standard particle, HFA) | 200–400 | | >400 |
| **Children 6–11 years – see notes above (for children 5 years and younger.** | | | |
| Beclometasone dipropionate (pMDI, standard particle, HFA) | 100–200 | >200–400 | >400 |
| Beclometasone dipropionate (pMDI, extrafine particle, HFA) | 50–100 | >100–200 | >200 |
| Budesonide (DPI, or pMDI, standard particle, HFA) | 100–200 | >200–400 | >400 |
| Budesonide (nebules) | 250–500 | >500–1000 | >1000 |
| Ciclesonide (pMDI, extrafine particle*, HFA) | 80 | >80–160 | >160 |
| Fluticasone furoate (DPI) | 50 | | n.a. |
| Fluticasone propionate (DPI) | 50–100 | >100–200 | >200 |
| Fluticasone propionate (pMDI, standard particle, HFA) | 50–100 | >100–200 | >200 |
| Mometasone furoate (pMDI, standard particle, HFA) | 100 | | 200 |

# Annex 9. Medications and doses for GINA Track 1: anti-inflammatory reliever (AIR) therapy

**GINA Track 1 – general principles**

**In GINA Track 1, the reliever inhaler is low-dose ICS-formoterol**, with or without maintenance ICS-formoterol. This is the preferred treatment approach for adults and adolescents with asthma, because it reduces severe exacerbations across treatment steps compared with using a SABA reliever; it uses a single medication for both reliever and maintenance treatment (less confusing for patients); and the patient's treatment can be stepped up and down if needed without changing the medication or inhaler device. This cannot be done with any other ICS-LABA. ICS-formoterol can also be used before exercise and before allergen exposure.

Low-dose ICS-formoterol is called an **anti-inflammatory reliever (AIR)** because it relieves symptoms and reduces inflammation. AIR with ICS-formoterol significantly reduces the risk of severe exacerbations across treatment steps compared with using a SABA reliever, with similar symptom control, lung function and adverse effects.

**Steps 1–2 (AIR-only)**: low-dose ICS-formoterol is used as needed for symptom relief without any maintenance treatment. It reduces the risk of severe exacerbations and ED visits/hospitalizations by 65% compared with SABAalone, and reduces ED visits/hospitalizations by 37%, compared with daily ICS plus as-needed SABA. Starting treatment with as-needed ICS-formoterol avoids training patients to regard SABA as their main asthma treatment.

**Steps 3–5 (MART)**: maintenance-and-reliever therapy with ICS-formoterol reduces the risk of severe exacerbations by 32% compared with the same dose of ICS-LABA, by 23% compared with a higher dose of ICS-LABA, and by 17% compared with usual care. MART is also an option for children 6–11 years in Steps 3–4.

**Asthma action plan:** Simple action plans for AIR-only and MART are available online.

**Which medications can be used in GINA Track 1, and how often?**

Most evidence for MART, and all evidence for AIR-only, is with budesonide-formoterol DPI, usually 200/6 mcg metered dose (160/4.5 mcg delivered dose) for adults/adolescents, and 100/6 mcg (80/4.5 mcg delivered dose) for MART in children 6–11 years. Beclometasone dipropionate (BDP)-formoterol 100/6 mcg (84.6/5.0) is also effective for MART in adults. Other low-dose combination ICS-formoterol products may be suitable but have not been studied.

For as-needed use, patients should take either 1 or 2 inhalations (based on the formulation; see below and next page) whenever needed for symptom relief, or before exercise or allergen exposure, instead of a SABA reliever.

Patients do not need to wait a certain number of hours before taking more reliever doses (unlike SABA), but in a single day, they should not take more than the maximum total number of inhalations shown below and over (total as-needed plus maintenance doses, if used). Most patients need far less than this.

| Age | Inhalers: mcg/inhalation metered dose [delivered dose] and maximum in any day | Dosing frequency by age group and treatment step |
|---|---|---|
| **6–11 Years** | Budesonide-formoterol 100/6 **DPI** *[80/4.5]* (maximum total 8 inhalations in any day) | Step 1–2 AIR-only: no evidence to date<br>Step 3 MART: 1 inhalation once daily plus 1 as needed<br>Step 4 MART: 1 inhalation twice daily plus 1 as needed Step 5 MART: not recommended |
| **12–17 Years** | Budesonide-formoterol 200/6 **DPI** [*160/4.5*] (maximum total 12 inhalations in any day) | Step 1–2 (AIR-only): 1 inhalation as needed<br>Step 3 MART: 1 inhalation twice (or once) daily plus 1 as needed<br>Step 4 MART: 2 inhalations twice daily plus 1 as needed<br>Step 5 MART: 2 inhalations twice daily plus 1 as needed |
| **≥18 Years** | Budesonide-formoterol 200/6 *[160/4.5]* **DPI**or BDP-formoterol 100/6 **pMDI** or **DPI** (maximum total 12 inhalations in any day†) | Step 1–2 (AIR-only): 1 inhalation as needed†<br>Step 3 MART: 1 inhalation twice (or once) daily plus 1 as needed<br>Step 4 MART: 2 inhalations twice daily plus 1 as needed<br>Step 5 MART: 2 inhalations twice daily plus 1 as needed |

For beclometasone (BDP)-formoterol, the maximum total dose in any day should be 12 inhalations.

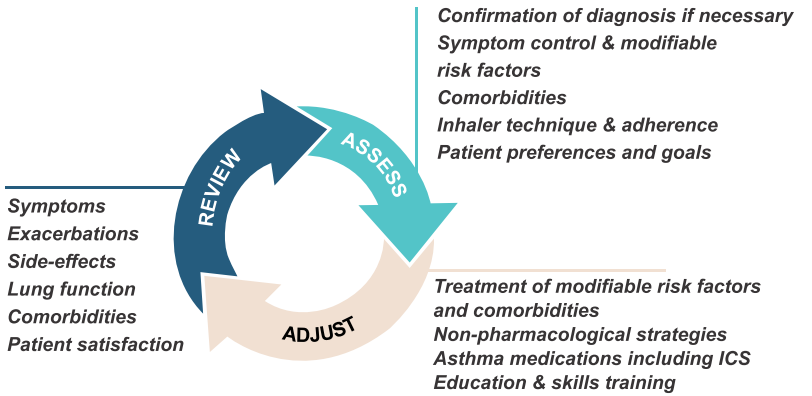
## Annex 10 (continued). Medications and doses for GINA Track 1 anti-inflammatory reliever (AIR) therapy

| Medications: mcg/inhalation metered dose [delivered dose] (maximum total inhalations in any day*) | Dosing frequency for ICS-formoterol formulations suitable for AIR therapy, by age group and treatment step |
|---|---|
| **Children 6–11 years** | |
| Budesonide-formoterol **DPI** 100/6 *[80/4.5]* (maximum total 8 inhalations in any day*) | Step 1–2 AIR-only: no evidence to date<br>Step 3 MART: 1 inhalation once daily plus 1 as needed Step 4 MART: 1 inhalation twice daily plus 1 as needed Step 5 MART: not recommended |
| Budesonide-formoterol **pMDI** 50/3 *[40/2.25]* (maximum total 16 inhalations in any day*) | Step 1–2 AIR-only: no evidence to date<br>Step 3 MART: 2 inhalations once daily plus 2 as needed<br>Step 4 MART: 2 inhalations twice daily plus 2 as needed Step 5 MART: not recommended |
| **Adolescents 12–17 years** | |
| Budesonide-formoterol **DPI** 200/6 *[160/4.5]* (maximum total 12 inhalations in any day*) | Step 1–2 (AIR-only): 1 inhalation as needed<br>Step 3 MART: 1 inhalation twice (or once) daily plus 1 as needed Step 4 MART: 2 inhalations twice daily plus 1 as needed<br>Step 5 MART: 2 inhalations twice daily plus 1 as needed |
| Budesonide-formoterol **pMDI** 200/6 *[160/4.5]* (maximum total 12 inhalations in any day*) | Step 1–2 (AIR-only): 1 inhalation as needed<br>Step 3 MART: 1 inhalation twice (or once) daily plus 1 as needed Step 4 MART: 2 inhalations twice daily plus 1 as needed<br>Step 5 MART: 2 inhalations twice daily plus 1 as needed |
| Budesonide-formoterol **pMDI** 100/3 *[80/2.25]* (maximum total 24 inhalations in any day*) | Step 1–2 (AIR-only): 2 inhalations as needed<br>Step 3 MART: 2 inhalations twice (or once) daily plus 2 as needed Step 4 MART: 4 inhalations twice daily plus 2 as needed<br>Step 5 MART: 4 inhalations twice daily plus 2 as needed |
| **Adults 18 years and older** | |
| Budesonide-formoterol **DPI** 200/6 *[160/4.5]* (maximum total 12 inhalations in any day*) | Step 1–2 (AIR-only): 1 inhalation as needed<br>Step 3 MART: 1 inhalation twice (or once) daily plus 1 as needed Step 4 MART: 2 inhalations twice daily plus 1 as needed<br>Step 5 MART: 2 inhalations twice daily plus 1 as needed |
| Budesonide-formoterol **pMDI** 200/6 *[160/4.5]* (maximum total 12 inhalations in any day*) | Step 1–2 (AIR-only): 1 inhalation as needed<br>Step 3 MART: 1 inhalation twice (or once) daily plus 1 as needed Step 4 MART: 2 inhalations twice daily plus 1 as needed<br>Step 5 MART: 2 inhalations twice daily plus 1 as needed |
| Budesonide-formoterol **pMDI** 100/3 *[80/2.25]* (maximum total 24 inhalations in any day*) | Step 1–2 (AIR-only): 2 inhalations as needed<br>Step 3 MART: 2 inhalations twice (or once) daily plus 2 as needed Step 4 MART: 4 inhalations twice daily plus 2 as needed<br>Step 5 MART: 4 inhalations twice daily plus 2 as needed |
| Beclometasone-formoterol **pMDI** or **DPI** 100/6 (GINA suggests maximum total 12 inhalations in any day*) | Step 1–2 (AIR-only): 1 inhalation as needed<br>Step 3 MART: 1 inhalation twice (or once) daily plus 1 as needed Step 4 MART: 2 inhalations twice daily plus 1 as needed<br>Step 5 MART: 2 inhalations twice daily plus 1 as needed |

# Annex 11. Personalized management for adults and adolescents to control symptoms and minimize future risk

## GINA 2024 - Adults & adolescents 12+ years

**Personalized asthma management**
Assess, Adjust, Review
for individual patient needs




**TRACK 1: PREFERRED CONTROLLER and RELIEVER**
Using ICS-formoterol as the reliever* reduces the risk of exacerbations compared with using a SABA reliever, and is a simpler regimen

| STEPS 1 - 2 | STEP 3 | STEP 4 | STEP 5 |
|---|---|---|---|
| As-needed-only low dose ICS-formoterol | Low dose maintenance ICS-formoterol | Medium dose maintenance ICS-formoterol | Add-on LAMA<br>Refer for assessment of phenotype. Consider high dose maintenance ICS-formoterol, ± anti-IgE, anti-IL5/5R, anti-IL4Rα, anti-TSLP |

**RELIEVER: As-needed low-dose ICS-formoterol***

See GINA severe asthma guide

**TRACK 2: Alternative CONTROLLER and RELIEVER**
Before considering a regimen with SABA reliever, check if the patient is likely to adhere to daily controller treatment

| STEP 1 | STEP 2 | STEP 3 | STEP 4 | STEP 5 |
|---|---|---|---|---|
| Take ICS whenever SABA taken* | Low dose maintenance ICS | Low dose maintenance ICS-LABA | Medium/high dose maintenance ICS- LABA | Add-on LAMA<br>Refer for assessment of phenotype. Consider high dose maintenance ICS- LABA + anti-IgE, anti-IL5/5R, anti-IL4Rα, anti-TSLP |

**RELIEVER: As-needed ICS-SABA*, or As-needed SABA**

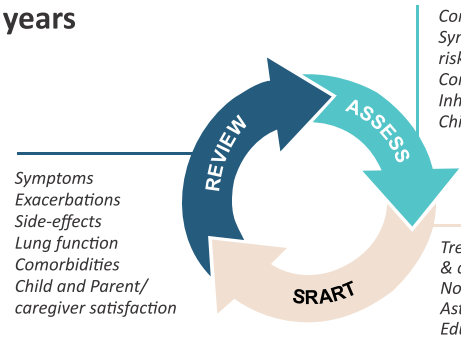
*Other controller options (limited indication, or less evidence for efficacy or safety )*

| | | | |
|---|---|---|---|
| *Low dose ICS whenever SABA taken*, Or daily LTRA• or add HDM SLIT* | *Mediam dose ICS, or add LTRA•, or add HDM SLIT* | *Add LAMA or add LTRA• or add HDM SLIT* **or switch to high dose ICS-only** | *Add azithromycin(adults) or add LTRA•,As last resort consider adding low dose OCS but consider side-effects* |

*Anti-inflammatory reliever,•If prescribe LTRA, advice patient/caregiver about risk of neuropsychiatric adverse effects.

# Annex 12 Personalized management for children 6–11 years to control symptoms and minimize future risk

## GINA 2024 - Children 6-11 years




*Confirmation of diagnosis, if necessary,*
*Symptom control & modifiable risk factors*
*Comorbidities*
*Inhaler technique &adherence*
*Child and parent/caregiver preferences and goals*

*Symptoms*
*Exacerbations*
*Side-effects*
*Lung function*
*Comorbidities*
*Child and Parent/ caregiver satisfaction*

*Treatment of modifiable risk factors & comorbidities*
*Non-pharmacological strategies*
*Asthma medications including ICS*
*Education & skills training*

**Asthma medication options:**
Adjust treament up and down for individual child´s needs

| | STEP 1 | STEP 2 | STEP 3 | STEP 4 | STEP 5 |
|---|---|---|---|---|---|
| **PREFERRED CONTROLLER** to prevent exacerbations and control symptoms | Low dose ICS taken whenever SABA taken* | Daily low dose inhaled corticosteroid (ICS) (see table of ICS dose ranges for children) | Low dose ICS-LABA, or medium dose ICS, OR very low dose ICS-formoterol maintenance and reliever (MART) | Refer for expert advice, OR medium dose ICS-LABA OR low dose ICS-formoterol maintenance and reliever therapy (MART) | Refer for phenotypic assessment ±higher dose ICS –LABA or add on therapy, e.g anti-IgE, anti-IL4Rα, anti-IL5 |
| **Other controller options (limited indications or less evidence for efficacy or safety)** | | *Daily leukotriene receptor antagonist (LTRA•), or low dose ICS taken whenever SABA taken** | *Low dose ICS + LTRA•* | *Add tiotropium or add LTRA•* | As last resort, consider add-on low dose OCS, but consider side-effects |
| **RELIEVER** | AS needed SABA (or ICS-formoterol reliever* in MART in Steps 3 and 4) | | | | |

تصميم وطبع
مطابع دائرة العيادات الطبية الشعبية
بغداد